HOPF INVARIANTS OF POLYNOMIAL MAPS AND VILLARCEAU SPHERES

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Abstract. We prove that the stereographic images of the fibres of the complex, quaternionic and octonionic Hopf fibrations that lie on a product of two round spheres are the Villarceau spheres of a tube, which gives Villarceau's theorem for every ring torus and its analogues in dimensions four and eight. We prove that the Hopf invariant of a holomorphic polynomial map from the three-sphere to the two-sphere is the number of common zeros of its two coordinates in the open four-ball, counted with multiplicity. For quaternionic and octonionic maps whose coordinates are one-sided polynomials in separate variables, we prove that the Hopf invariant is a sum of positive contributions of the zero points and zero spheres in the ball, read off from two real polynomials; every value from zero to the product of the degrees occurs, the formula extends to triangular maps, and positivity fails for general polynomial maps, which realise every integer. Along the way we prove that the three Hopf maps are Riemannian submersions and sphere bundles of vector bundles with Hopf invariant one, and we derive the homotopy groups of the spheres of dimensions two, four and eight from those of the spheres of dimensions three and seven.

1. Introduction

1.1. **The objects.** A continuous map between spheres is determined up to homotopy by remarkably little, and yet the groups $\pi_i(S^n)$ of such maps are among the hardest invariants in mathematics to compute. Below the dimension of the target nothing happens: $\pi_i(S^n) = 0$ for $i < n$. In the dimension of the target the degree is a complete invariant, $\pi_n(S^n) \cong \mathbb{Z}$. The first surprise is $\pi_3(S^2) \cong \mathbb{Z}$, found by Hopf in 1931 [Hop31]: a map from a three-sphere onto a two-sphere can wrap in infinitely many essentially different ways, and the generator is the map

$$h\colon S^3 \to S^2, \qquad h(z,w) = \big(2z\bar{w},\ |z|^2 - |w|^2\big) \in \mathbb{C}\times\mathbb{R} = \mathbb{R}^3, \tag{1.1}$$

whose fibres are the great circles $\{(\lambda z, \lambda w) : |\lambda| = 1\}$. Any two of these circles are linked once. Seen in $\mathbb{R}^3$ through stereographic projection, the fibres lying over a circle of latitude sweep out a torus of revolution, and on that torus they are not meridians, not parallels, but the oblique circles discovered by Yvon Villarceau in 1848 [Vil48]: the circles cut from a ring torus by a plane through its centre that touches it at two points.

Hopf's construction has exactly two more analogues, built from the quaternions and the octonions, with fibres S^3 and S^7 [Hop35]. Adams' theorem on the Hopf invariant [Ada60] says that there are no others, and that no sphere bundle $S^{d-1} \to S^{2d-1} \to S^d$ exists unless $d \in \{1, 2, 4, 8\}$.

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This paper gives a complete account of these objects with proofs, and it adds results that make the connection between homotopy and geometry precise. We have aimed at a text in which every statement is either proved here or is a published theorem quoted with its source; the deep inputs that are quoted, and not reproved, are listed in Section 1.3. Nothing below depends on a conjecture. The text replaces the earlier versions [BSS22], [Bha26a] of this work, which were expository, corrects their errors, and adds Theorems D and E, Proposition 8.3, the results of Section 8 on Villarceau spheres, the Riemannian submersion of Proposition 7.5 for every division algebra, and the local formula for the Hopf invariant of split and triangular quaternionic and octonionic polynomial maps in Section 6.3 (Theorems 6.15, 6.18 and Corollary 6.17).

1.2. **Results.** Throughout, $\mathbb{K}$ is one of the four normed division algebras $\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$ and $d = \dim_{\mathbb{R}} \mathbb{K} \in \{1, 2, 4, 8\}$. We write $\bar{x}$ for the conjugate of $x \in \mathbb{K}$ and $|x|^2 = x\bar{x}$. The *Hopf map* of $\mathbb{K}$ is

$$h_{\mathbb{K}} \colon S^{2d-1} \to S^d, \qquad h_{\mathbb{K}}(a, b) = \big(2a\bar{b},\ |a|^2 - |b|^2\big) \in \mathbb{K} \times \mathbb{R} = \mathbb{R}^{d+1}, \tag{1.2}$$

where S^{2d-1} is the unit sphere of $\mathbb{K}^2 = \mathbb{R}^{2d}$. For $x \in \mathbb{K}$ let

$$L_x = \{(xc, c) : c \in \mathbb{K}\} \subset \mathbb{K}^2, \qquad L_\infty = \mathbb{K} \times \{0\}.$$

These are real linear subspaces of dimension d, and we view the family $\{L_x\}$ as parametrised by the one-point compactification $\mathbb{K} \cup \{\infty\} \cong S^d$.

Theorem A (Hopf bundles)**.** *For each $\mathbb{K}$ the subspaces L_x, $x \in \mathbb{K} \cup \{\infty\}$, form a smooth real vector bundle $L_{\mathbb{K}}$ of rank d over S^d, a subbundle of the trivial bundle $S^d \times \mathbb{K}^2$. Every nonzero vector of $\mathbb{K}^2$ lies in exactly one L_x, so the unit sphere bundle of $L_{\mathbb{K}}$ is S^{2d-1} itself, and its projection is the Hopf map* (1.2)*. The fibres of $h_{\mathbb{K}}$ are great $(d-1)$-spheres, and the bundle is obtained by gluing two trivial bundles over the hemispheres along the map $x \mapsto (c \mapsto xc)$ of the equator S^{d-1} into $\mathrm{GL}_d(\mathbb{R})$.*

For $\mathbb{K} = \mathbb{O}$ the octonions are not associative, and the usual definition of the octonionic projective line requires care; Theorem A replaces it by an explicit vector bundle. The proof uses only Artin's theorem that two octonions generate an associative subalgebra.

Theorem B (Hopf invariant)**.** *Let $d \in \{2, 4, 8\}$.*

(i) *The Hopf invariant of $h_{\mathbb{K}}$ is ± 1.*
(ii) *For every integer $m \geq 1$ the power map $x \mapsto x^m$ of the unit sphere $S^{d-1} \subset \mathbb{K}$ has degree m, and the map $g_m(a, b) = (a^m, b)/|(a^m, b)|$ of S^{2d-1} has degree m.*
(iii) *The composite $h_{\mathbb{K}} \circ g_m$ represents $m[h_{\mathbb{K}}]$ and has Hopf invariant $\pm m$; composing with $(a, b) \mapsto (\bar{a}, b)$ gives $-m$. In particular $\mathrm{H} \colon \pi_{2d-1}(S^d) \to \mathbb{Z}$ is onto, and each value is attained by a map whose coordinates are polynomials in the real coordinates of $\mathbb{K}^2$ divided by a norm.*

Theorem C (Splitting)**.** *For $d \in \{2, 4, 8\}$ and every $i \geq 2$,*

$$\pi_i(S^d) \ \cong\ \pi_i(S^{2d-1}) \oplus \pi_{i-1}(S^{d-1}),$$

where the first summand is embedded by $(h_{\mathbb{K}})_*$. In particular $\pi_i(S^2) \cong \pi_i(S^3)$ for $i \geq 3$, $\pi_7(S^4) \cong \mathbb{Z} \oplus \mathbb{Z}/12$ and $\pi_{15}(S^8) \cong \mathbb{Z} \oplus \mathbb{Z}/120$, with the infinite cyclic summands generated by the Hopf maps and detected by the Hopf invariant.

The splitting determines the columns $n = 2, 4, 8$ of the table of $\pi_{n+k}(S^n)$ from the columns $n = 3, 7$ and from $\pi_*(S^{15})$; Figure 2 and Table 1 show which entries are derived in this way.

The next result concerns maps $S^3 \to S^2$ given by polynomials. For $P, Q \in \mathbb{C}[z,w]$ without common zero on S^3 put

$$[P:Q]\colon S^3 \to S^2, \qquad [P:Q](u) = h\big(P(u),Q(u)\big)\big/\big(|P(u)|^2+|Q(u)|^2\big),$$

that is, the point $[P(u):Q(u)]$ of $\mathbb{CP}^1 = S^2$.

Theorem D (Holomorphic maps)**.** *Let $P, Q \in \mathbb{C}[z,w]$ have no common zero on S^3, and orient so that $\mathrm{H}(h) = 1$. Then P and Q have finitely many common zeros in the open unit ball $B^4 \subset \mathbb{C}^2$, and*

$$\mathrm{H}\big([P:Q]\big) = \sum_{u\in B^4,\ P(u)=Q(u)=0} I_u(P,Q) \ \in\ \big[0,\ \deg P \cdot \deg Q\big],$$

where $I_u(P,Q) = \dim_{\mathbb{C}} \mathcal{O}_u/(P,Q)$ is the intersection multiplicity. Both bounds are attained, the upper one by $[z^p : w^q]$, whose fibres are torus links of type (p,q) linked pq times. A holomorphic map $[P:Q]$ is null-homotopic exactly when P and Q have no common zero in B^4, and every class in $\pi_3(S^2) \cong \mathbb{Z}$ is represented by a map $[P:Q]$ or by its composite with complex conjugation.

Theorem D turns the Hopf invariant, a homotopy invariant, into an algebraic count, and turns the linking of fibres into an intersection number of plane curves inside the ball.

The next two results are geometric. Let $0 < a < 1$ and $b = \sqrt{1-a^2}$, and let $\varepsilon \in \mathbb{K}$ be a unit with $\varepsilon \perp 1$. Stereographic projection from $N = (0,\varepsilon)$ identifies $S^{2d-1} \setminus \{N\}$ with $\mathbb{K} \times \varepsilon^{\perp} = \mathbb{R}^d \times \mathbb{R}^{d-1}$.

Theorem E (Villarceau spheres)**.** *Let $d \in \{2,4,8\}$ and put $R = 1/a$, $r = b/a$, so that $R^2 - r^2 = 1$. Stereographic projection maps the product $S^{d-1}(a) \times S^{d-1}(b)$ onto the tube*

$$T_{R,r} = \big\{(x,y) \in \mathbb{R}^d \times \mathbb{R}^{d-1} : (|x|-R)^2 + |y|^2 = r^2\big\}$$

of radius r about the round sphere of radius R in $\mathbb{R}^d \times 0$. It maps every fibre of $h_{\mathbb{K}}$ contained in $S^{d-1}(a) \times S^{d-1}(b)$ onto a round $(d-1)$-sphere of radius R, centred at a point at distance r from the origin and lying in a d-dimensional linear subspace; these spheres partition $T_{R,r}$. For $d = 2$ the tube is a ring torus with $R^2 - r^2 = 1$, the images of the fibres of h form one family of Villarceau circles, and the images of the fibres of $(z,w) \mapsto h(\bar z, w)$ form the other.

Corollary 1.1 (Villarceau's theorem)**.** *Let T be a ring torus with radii $R > r > 0$. Every plane through the centre of T tangent to T at two points meets T in two circles of radius R, and each of the two families of such circles is the image of a family of Hopf fibres under stereographic projection followed by the similarity of ratio $\sqrt{R^2 - r^2}$.*

Theorem F (Metric structure)**.** *For $\mathbb{K} = \mathbb{C}, \mathbb{H}, \mathbb{O}$ the Hopf map is a Riemannian submersion from the unit sphere S^{2d-1} onto the sphere $S^d(\frac{1}{2})$ of radius $\frac{1}{2}$. The fibres are parallel great spheres: the distance between the fibres over $p, q \in S^d$ is half the angle between p and q. For $\mathbb{K} = \mathbb{C}$ the fibres over a circle of latitude form a flat torus $S^1(a) \times S^1(b)$ on which they are the closed geodesics of slope b/a, and the unit tangent bundle of S^2 is the quotient of the Hopf bundle by ± 1, a circle bundle over S^2 with total space $\mathbb{RP}^3$.*

We return to polynomial maps. Over $\mathbb{H}$ and $\mathbb{O}$ there is no intersection theory, and Theorem D survives only in a restricted form. A *one-sided polynomial* of degree m over $\mathbb{K}$ is $P(q) = \sum_{k=0}^{m} q^k p_k$ with $p_k \in \mathbb{K}$, $p_m \neq 0$; its *norm polynomial* is the real polynomial $S_P(z) = \sum_{i,j} \langle p_i, p_j\rangle z^{i+j}$ of degree $2m$.

Theorem G (Quaternionic and octonionic maps)**.** *Let $\mathbb{K} = \mathbb{H}$ or $\mathbb{O}$, let P, Q be one-sided polynomials of degrees $m, n \geq 1$, and suppose that $F(a,b) = (P(a), Q(b))$ has no zero on S^{2d-1}.*

(i) *The zero set of P consists of finitely many points and round $(d-2)$-spheres, one for each root of S_P in the closed upper half-plane; the component of a root of multiplicity ℓ has index ℓ, or $\ell/2$ if the root is real, and the indices add up to m.*
(ii) $\mathrm{H}(h_{\mathbb{K}} \circ F/|F|) = \mathrm{H}(h_{\mathbb{K}}) \sum \mu_\alpha \nu_\beta$, *the sum over the pairs of zero components of P and Q, of radii r_α, s_β and indices μ_α, ν_β, with $r_\alpha^2 + s_\beta^2 < 1$. Every value $0, 1, \dots, mn$ of the sum occurs.*
(iii) *The same bounds $0 \le \deg(F/|F|) \le mn$ hold for the triangular maps $(P(a), Q(b) + R(a))$ with R an arbitrary polynomial, while polynomial maps with constants between the variable factors attain every integer degree.*

The proofs are in Section 6.3. The positivity behind (i) and (ii) is a Jacobian identity, $\det DP = |B|^{d-4}(s^2+t^2) \ge 0$ (Proposition 6.11), which replaces the complex identity $\det_{\mathbb{R}} DF = |\det_{\mathbb{C}} DF|^2$.

Sections 9 and 10 place these results in their wider setting: the stable stems through dimension 19, the image of the J-homomorphism and its Bernoulli denominators, the Kervaire invariant problem, and the torus fibrations of the Strominger–Yau–Zaslow picture, of which the Clifford tori of the Hopf fibration are the simplest example. We state precisely what is known there, prove what is elementary, and separate both from the conjectural.

1.3. **Inputs taken from the literature.** The following theorems are used without proof. Each is a published result with a complete proof in the cited source.

(I1) Cellular approximation, the long exact sequence of a fibration, the Hurewicz theorem and the Freudenthal suspension theorem [Hat02, Chapter 4].
(I2) Poincaré duality for closed oriented manifolds [Hat02, Section 3.3].
(I3) Artin's theorem: the subalgebra of $\mathbb{O}$ generated by any two elements is associative [Sch66, Chapter III], [Bae02].
(I4) Adams' theorem: an element of Hopf invariant one in $\pi_{2n-1}(S^n)$ exists only for $n = 1, 2, 4, 8$ [Ada60], [AA66].
(I5) The degree of the boundary map of a vector field with isolated zeros equals the sum of their indices [Mil65, Section 6]; the index of an isolated zero of a holomorphic map germ $(\mathbb{C}^2, u) \to (\mathbb{C}^2, 0)$ is its intersection multiplicity [GH78, Chapter 5], [AGV85, Chapter 5]; Bézout's theorem [Ful69, Chapter 5]; an irreducible complex algebraic curve is connected in the classical topology [Mum76, Corollary (4.16)].
(I6) The values of $\pi_{n+k}(S^n)$ for $k \le 7$ and of the stable stems for $k \le 19$ [Tod62], [Rav86], [IWX23]; the image of J [Ada66], [Qui71], [Sul74], [BG75]; the Kervaire invariant results [Bro69], [HHR16].
(I7) Hopf's description of the Hopf invariant of a map $S^3 \to S^2$ as the linking number of two fibres [Hop31], [BT82, Section 17].
(I8) A real algebraic subset of $\mathbb{R}^N$ has finitely many connected components [BCR98, Theorem 2.4.4].

Items (I4) and (I6) are deep, and their proofs are not reproduced. The rest of the text is self-contained modulo standard algebraic topology.

1.4. **Plan.** Section 2 fixes conventions and proves the elementary facts about homotopy groups of spheres that are needed, together with a splitting lemma for fibrations whose fibre inclusion is null-homotopic. Section 3 constructs the Hopf bundles and proves Theorem A. Section 4 treats the Hopf invariant and proves Theorem B; Section 5 proves Theorem C and derives the table of unstable groups. Section 6 is about linking numbers and proves Theorem D, together with its analogues for split and triangular quaternionic and octonionic maps (Theorem G), and the failure of positivity for general ones. Section 7 proves Theorem F, and Section 8 proves Theorem E

and Villarceau's theorem. The appendices collect the octonion identities, the algebra behind Villarceau's theorem, the tables of stable stems and the numerical checks.

1.5. **Conventions.** Spheres are $S^n = \{x \in \mathbb{R}^{n+1} : |x| = 1\}$ and $S^n(\rho)$ is the sphere of radius ρ. Homotopy groups are based, and we drop basepoints from the notation when the space is path connected. A map $f\colon S^n \to S^n$ has a degree $\deg f \in \mathbb{Z}$, defined by $f_*[S^n] = (\deg f)[S^n]$ in $H_n(S^n;\mathbb{Z})$. We write $\mathbb{Z}/m$ for the cyclic group of order m, and in tables a cyclic group is denoted by its order. The words *fibration* and *fibre bundle* are used in their standard senses [Hat02, Section 4.2]; all fibre bundles below are locally trivial over spheres and therefore Hurewicz fibrations.

2. Homotopy groups and fibrations

This section collects the facts from homotopy theory used later. Most are standard; we give proofs where the argument is short or where the precise form matters, and we prove one splitting lemma, Lemma 2.8, on which Section 5 rests.

2.1. **Homotopy groups.** Let X be a space with basepoint x_0. For $i \geq 1$ the group $\pi_i(X, x_0)$ is the set of homotopy classes of maps $f\colon (I^i, \partial I^i) \to (X, x_0)$, the homotopies being through maps of the same kind, with the product

$$(f+g)(t_1,\dots,t_i) = \begin{cases} f(2t_1, t_2, \dots, t_i), & t_1 \leq \tfrac{1}{2}, \\ g(2t_1 - 1, t_2, \dots, t_i), & t_1 \geq \tfrac{1}{2}. \end{cases}$$

It is abelian for $i \geq 2$.[1] Collapsing ∂I^i identifies $\pi_i(X, x_0)$ with the set of based homotopy classes of based maps $(S^i, s_0) \to (X, x_0)$. For a pair (X, A) with $x_0 \in A$ and $i \geq 1$, the relative set $\pi_i(X, A, x_0)$ consists of homotopy classes of maps $(D^i, S^{i-1}, s_0) \to (X, A, x_0)$; it is a group for $i \geq 2$, and there is an exact sequence

$$\cdots \to \pi_i(A) \to \pi_i(X) \to \pi_i(X, A) \xrightarrow{\partial} \pi_{i-1}(A) \to \cdots \to \pi_0(X), \tag{2.1}$$

in which ∂ restricts a map of D^i to its boundary [Hat02, Theorem 4.3].

A *CW complex* is a space X with a filtration $X^0 \subset X^1 \subset \cdots$ by skeleta, where X^0 is discrete and X^n is obtained from X^{n-1} by attaching n-cells along maps $S^{n-1} \to X^{n-1}$, and $X = \bigcup_n X^n$ has the weak topology. The closure of each cell then meets only finitely many cells.[2] The sphere S^n has a CW structure with one 0-cell s_0 and one n-cell attached by the constant map. A map $f\colon X \to Y$ of CW complexes is *cellular* if $f(X^n) \subset Y^n$ for all n.

Theorem 2.1 (Cellular approximation [Hat02, Theorem 4.8])**.** *Every map of CW complexes is homotopic to a cellular map, by a homotopy that is stationary on any subcomplex on which the map is already cellular.*

Proposition 2.2. *$\pi_i(S^n) = 0$ for $0 \leq i < n$.*

Proof. Give S^i and S^n the CW structures with one 0-cell each, at the basepoints. A based map $f\colon S^i \to S^n$ is cellular on the subcomplex $\{s_0\}$, so by Theorem 2.1 it is homotopic relative to s_0 to a cellular map. A cellular map sends $S^i = (S^i)^i$ into the i-skeleton of S^n, which is the point s_0 because $i < n$. So f is null-homotopic relative to the basepoint. □

[1] The two halves of the cube can be shrunk and slid around one another in the second coordinate; for $i = 1$ there is no second coordinate, and π_1 need not be abelian.

[2] This is the "closure-finite" condition that gives CW complexes the letter C. It concerns the closure of a cell, which lies in a finite subcomplex; a cell may well meet the closures of infinitely many others.

Proposition 2.3. *For $n \geq 1$ the degree defines an isomorphism $\pi_n(S^n) \to \mathbb{Z}$ that sends the identity to 1. Two maps $S^n \to S^n$ are freely homotopic if and only if they have the same degree.*

Proof. For $n = 1$ this is the computation of $\pi_1(S^1)$ by covering space theory. For $n \geq 2$ the sphere is $(n-1)$-connected by Proposition 2.2, and the Hurewicz homomorphism $\pi_n(S^n) \to H_n(S^n; \mathbb{Z}) \cong \mathbb{Z}$, which sends $[f]$ to $f_*[S^n] = (\deg f)[S^n]$, is an isomorphism [Hat02, Theorem 4.32]. For the second statement note that $\pi_1(S^n)$ acts trivially on $\pi_n(S^n)$ (for $n \geq 2$ it is the trivial group, and for $n = 1$ the group is abelian), so free homotopy classes of maps $S^n \to S^n$ coincide with based ones [Hat02, Section 4.A]. □

The antipodal map of S^n is the composite of $n + 1$ reflections, each of degree -1, so it has degree $(-1)^{n+1}$.

Lemma 2.4. *Let X be a space, $i \geq 1$, $\alpha \in \pi_i(X)$, and let $g\colon S^i \to S^i$ be a based map of degree m. Then $\alpha \circ g = m\alpha$.*

Proof. For a fixed map $f\colon S^i \to X$, composition $[g] \mapsto [f \circ g]$ is a homomorphism $f_*\colon \pi_i(S^i) \to \pi_i(X)$, because the sum in π_i is formed in the source coordinate. By Proposition 2.3 the class of g is $m[\mathrm{id}]$, so $[f \circ g] = f_*(m[\mathrm{id}]) = m[f]$. □

2.2. **Fibrations and the long exact sequence.** A map $p\colon E \to B$ has the *homotopy lifting property* for a space Y if for every homotopy $F\colon Y \times I \to B$ and every lift $\tilde{F}_0\colon Y \to E$ of F_0 there is a homotopy $\tilde{F}$ lifting F and starting at $\tilde{F}_0$. It is a *Serre fibration* if it has this property for all cubes $Y = I^k$, and a *Hurewicz fibration* if it has it for all spaces. Every fibre bundle is a Serre fibration, and a fibre bundle over a paracompact base is a Hurewicz fibration [Hat02, Proposition 4.48 and the remarks after it].[3]

Theorem 2.5 ([Hat02, Theorem 4.41])**.** *Let $p\colon E \to B$ be a Serre fibration, $b_0 \in B$, $F = p^{-1}(b_0)$ and $x_0 \in F$. Then $p_*\colon \pi_i(E, F, x_0) \to \pi_i(B, b_0)$ is an isomorphism for $i \geq 1$. Consequently, with ∂ the composite of p_*^{-1} and the boundary map of (2.1), there is an exact sequence*

$$\cdots \to \pi_i(F) \xrightarrow{i_*} \pi_i(E) \xrightarrow{p_*} \pi_i(B) \xrightarrow{\partial} \pi_{i-1}(F) \to \cdots \to \pi_0(E) \to \pi_0(B).$$

Concretely, ∂ is computed as follows.[4] Write $I^i = I^{i-1} \times I$ and let $f\colon (I^i, \partial I^i) \to (B, b_0)$ represent an element of $\pi_i(B)$. Regard f as a homotopy $f_t = f(\cdot, t)$ of maps $I^{i-1} \to B$ and lift it, starting at the constant map with value x_0 and keeping the faces $\partial I^{i-1} \times I$ at x_0 (this is possible by the lifting property for the pair $(I^{i-1} \times I, I^{i-1} \times 0 \cup \partial I^{i-1} \times I)$, which is homeomorphic to $(I^{i-1} \times I, I^{i-1} \times 0)$). The lift at time 1 is a map $(I^{i-1}, \partial I^{i-1}) \to (F, x_0)$, and its class is $\partial[f]$.

Example 2.6. For the universal covering $\mathbb{R} \to S^1$, with contractible total space and discrete fibre, the sequence gives $\pi_i(S^1) = 0$ for $i \geq 2$.

Example 2.7 (The real Hopf fibration)**.** The map $S^1 \to S^1$, $z \mapsto z^2$, is a two-sheeted covering with fibre S^0. In the notation (1.2) with $\mathbb{K} = \mathbb{R}$, the circle $S^1 \subset \mathbb{R}^2$ consists of the points (a, b) with $a^2 + b^2 = 1$, and $h_{\mathbb{R}}(a, b) = (2ab, a^2 - b^2)$ is the squaring map in the coordinate $a + ib$. It is the case $d = 1$ of Theorem A.

The following lemma is used three times, once for each of the Hopf bundles with $d = 2, 4, 8$.

[3] Covering spaces are fibre bundles with discrete fibre; for them the lift is unique, which is the path lifting property of covering space theory. The projection $B \times F \to B$ is the trivial bundle: a lift of F_t is $(F_t, \mathrm{pr}_F \circ \tilde{F}_0)$.

[4] This is the construction sketched in the earlier version of the work, made precise: the boundary of the lifted cube lies in the fibre, except for one face that is mapped to the basepoint.

Lemma 2.8 (Splitting)**.** *Let $p\colon E \to B$ be a Serre fibration with fibre $F = p^{-1}(b_0)$, all three spaces path connected, and suppose that the inclusion $j\colon F \to E$ is null-homotopic. Let $i \geq 2$ and assume that $\pi_{i-1}(F)$ is abelian. Then*

$$0 \to \pi_i(E) \xrightarrow{p_*} \pi_i(B) \xrightarrow{\partial} \pi_{i-1}(F) \to 0$$

is a split short exact sequence, so $\pi_i(B) \cong \pi_i(E) \oplus \pi_{i-1}(F)$. A splitting $\sigma\colon \pi_{i-1}(F) \to \pi_i(B)$ is given by (2.2) *below.*

Proof. Since j is null-homotopic, $j_* = 0$ on every homotopy group, and exactness of the sequence of Theorem 2.5 at $\pi_i(E)$ and at $\pi_{i-1}(F)$ shows that p_* is injective and ∂ is surjective. It remains to construct a homomorphism σ with $\partial \circ \sigma = \mathrm{id}$.

A null-homotopy of j is the same thing as an extension $G\colon CF \to E$ of j to the cone $CF = F \times I/F \times \{1\}$, where F is identified with $F \times \{0\}$. The cone is contractible, so the boundary map $\partial_C\colon \pi_i(CF, F, x_0) \to \pi_{i-1}(F, x_0)$ of the pair (CF, F) is an isomorphism for $i \geq 2$, by (2.1). Define

$$\sigma = p_* \circ G_* \circ \partial_C^{-1}\colon\ \pi_{i-1}(F) \to \pi_i(CF, F) \to \pi_i(E, F) \to \pi_i(B, b_0) = \pi_i(B). \tag{2.2}$$

Each of the three maps is a homomorphism of groups for $i \geq 2$, so σ is a homomorphism. By the definition of ∂ in Theorem 2.5,

$$\partial \circ \sigma = \partial_{(E,F)} \circ p_*^{-1} \circ p_* \circ G_* \circ \partial_C^{-1} = \partial_{(E,F)} \circ G_* \circ \partial_C^{-1} = j_* \circ \partial_C \circ \partial_C^{-1} = \mathrm{id},$$

where the third equality is the naturality of boundary maps for the map of pairs $G\colon (CF, F) \to (E, F)$, which restricts to the identity of F. □

Remark 2.9. In words: a class $\beta \in \pi_{i-1}(F)$ is represented by $\beta\colon S^{i-1} \to F$; the cone on β followed by G is a disc in E with boundary β in F, and $\sigma(\beta)$ is the image of that disc in B, a sphere because its boundary collapses to b_0.

2.3. **Suspension.** The reduced suspension of a based space (X, x_0) is $\Sigma X = X \times I/(X \times \partial I \cup x_0 \times I)$, and $\Sigma S^n \cong S^{n+1}$. A based map f induces $\Sigma f[x, t] = [f(x), t]$, and $f \mapsto \Sigma f$ induces the suspension homomorphism $E\colon \pi_i(S^n) \to \pi_{i+1}(S^{n+1})$.

Theorem 2.10 (Freudenthal [Fre38], [Hat02, Corollary 4.24])**.** *The suspension $E\colon \pi_i(S^n) \to \pi_{i+1}(S^{n+1})$ is an isomorphism for $i < 2n-1$ and is surjective for $i = 2n-1$.*

Equivalently, the adjoint inclusion $S^n \to \Omega S^{n+1}$ of S^n into the based loop space of S^{n+1} induces isomorphisms on π_i for $i < 2n-1$ and a surjection for $i = 2n-1$; it is $(2n-1)$-connected.[5] Hence for fixed k the groups $\pi_{n+k}(S^n)$ are independent of n once $n \geq k+2$; the common value is the *k-th stable stem* $\pi_k^s = \operatorname{colim}_n \pi_{n+k}(S^n)$. By Serre's theorem [Ser53] the groups $\pi_i(S^n)$ are finite except for $\pi_n(S^n) \cong \mathbb{Z}$ and, for n even, $\pi_{2n-1}(S^n)$, which is the direct sum of $\mathbb{Z}$ and a finite group. In particular π_k^s is finite for $k \geq 1$.

The path fibration $\Omega X \to PX \to X$, where PX is the contractible space of paths starting at x_0 and the map evaluates at the endpoint, gives $\pi_i(\Omega X) \cong \pi_{i+1}(X)$ by Theorem 2.5.

2.4. **Whitehead products.** Let $\iota_1, \iota_2\colon S^n \to S^n \vee S^n$ be the inclusions of the two wedge summands. The product $S^n \times S^n$ is obtained from $S^n \vee S^n$ by attaching one $2n$-cell along a map $w\colon S^{2n-1} \to S^n \vee S^n$, the attaching map of the top cell.[6] For $\alpha, \beta \in \pi_n(X)$ represented by f, g, the *Whitehead product* $[\alpha, \beta] \in \pi_{2n-1}(X)$ is the class of $(f \vee g) \circ w$. We need only the square $[\iota_n, \iota_n] \in \pi_{2n-1}(S^n)$ of the identity, the class of $\nabla \circ w$ with $\nabla\colon S^n \vee S^n \to S^n$ the fold map.

[5] It is this map, and not the inclusion of S^n as an equator of S^{n+1}, that is $(2n-1)$-connected; the equatorial inclusion $S^n \to S^{n+1}$ is only n-connected, since S^{n+1} is n-connected and $\pi_n(S^n) \to \pi_n(S^{n+1}) = 0$ is not injective.

[6] Write $S^n = D^n/\partial D^n$. Then $S^n \times S^n$ is the quotient of $D^n \times D^n$, whose boundary $\partial(D^n \times D^n) = \partial D^n \times D^n \cup D^n \times \partial D^n \cong S^{2n-1}$ is mapped onto $S^n \vee S^n$. This boundary map is w.

3. The Hopf bundles

This section proves Theorem A. The four cases are treated at once; the only place where the octonions need care is Lemma 3.1.

3.1. **The normed division algebras.** The algebras $\mathbb{R} \subset \mathbb{C} \subset \mathbb{H} \subset \mathbb{O}$ are obtained by the Cayley–Dickson construction. If A is a real algebra with unit and an involution $x \mapsto \bar{x}$, then $A \oplus A$ with

$$(a,b)(c,d) = \big(ac - \bar{d}b,\ da + b\bar{c}\big), \qquad \overline{(a,b)} = (\bar{a}, -b), \tag{3.1}$$

is again a real algebra with unit $(1,0)$ and involution. Starting from $\mathbb{R}$ with the trivial involution one obtains $\mathbb{C}$, $\mathbb{H}$ and $\mathbb{O}$, of dimensions $2, 4, 8$.[7] We collect the facts that are used, with references.

- (D1) $x\bar{x} = \bar{x}x = |x|^2$ is the Euclidean norm squared, and $|xy| = |x|\,|y|$ for all $x, y \in \mathbb{K}$ [Bae02, Section 2.2].
- (D2) $\overline{xy} = \bar{y}\,\bar{x}$, and $\bar{x} = 2\operatorname{Re} x - x$, where $\operatorname{Re} x = \frac{1}{2}(x + \bar{x}) \in \mathbb{R}$ is the real part [Bae02, Section 2.2].
- (D3) The Euclidean inner product is $\langle x, y\rangle = \operatorname{Re}(x\bar{y}) = \operatorname{Re}(\bar{x}y)$, and $\langle xa, b\rangle = \langle a, \bar{x}b\rangle$, $\langle ax, b\rangle = \langle a, b\bar{x}\rangle$ [Sch66, Chapter III].
- (D4) (Artin) Any two elements of $\mathbb{K}$ generate an associative subalgebra (input (I3); [Sch66, Theorem 3.1]).

For $\mathbb{K} = \mathbb{R}, \mathbb{C}, \mathbb{H}$ all four are immediate, since these algebras are associative. The identities (D1)–(D3) are checked by direct computation from (3.1) for $\mathbb{O}$, and they are verified numerically, together with Artin's theorem and the Moufang identities, by the programs of Appendix C.

Lemma 3.1. *Let $x, y \in \mathbb{K}$ with $y \neq 0$, and put $y^{-1} = \bar{y}/|y|^2$. Then*

$$(xy)y^{-1} = x, \qquad (xy^{-1})y = x, \qquad y^{-1}(yx) = x, \qquad (xy)\bar{y} = |y|^2 x, \qquad x(\bar{x}y) = |x|^2 y.$$

Proof. By (D2), $\bar{y} = 2\operatorname{Re} y - y$ lies in the subalgebra generated by 1 and y, so x, y, $\bar{y}$, y^{-1} and $\bar{x}$ all lie in the subalgebra A generated by x and y. By Artin's theorem A is associative, so $(xy)y^{-1} = x(yy^{-1}) = x$, and similarly for the other identities, using $y\bar{y} = |y|^2$ and $\bar{x}x = |x|^2$ from (D1). □

For $x \in \mathbb{K}$ let $\ell_x \colon \mathbb{K} \to \mathbb{K}$, $\ell_x(c) = xc$, be left multiplication. It is $\mathbb{R}$-linear, and by (D1) it multiplies lengths by $|x|$. Hence for $x \neq 0$ it is a similarity of ratio $|x|$, with inverse $\ell_{x^{-1}}$ by Lemma 3.1, and for $|x| = 1$ it is orthogonal. We obtain a smooth map

$$\tau \colon S^{d-1} \to \mathrm{O}(d), \qquad \tau(x) = \ell_x. \tag{3.2}$$

Remark 3.2. For $\mathbb{K} = \mathbb{R}, \mathbb{C}, \mathbb{H}$ the unit sphere S^{d-1} is a group and τ is a homomorphism, $\ell_x \ell_y = \ell_{xy}$. For $\mathbb{O}$ this fails: $\ell_x \ell_y = \ell_{xy}$ for all x, y would make $\mathbb{O}$ associative. The unit sphere S^7 is not a group, and no Lie group structure on S^7 exists at all, since the only spheres that are Lie groups are S^0, S^1 and S^3.[8] The construction below uses only the maps ℓ_x and Lemma 3.1, and it works uniformly.

[7] Doubling $\mathbb{O}$ once more gives the sedenions, which have zero divisors; this is one way to see that the list stops. The theorem that it stops is Hurwitz's: a finite-dimensional real algebra with unit and a multiplicative positive definite norm is one of $\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$ [Bae02, Section 2]. We do not need it, since Adams' theorem (I4) gives the stronger topological statement.

[8] A compact connected Lie group G of positive dimension has $H^3(G; \mathbb{R}) \neq 0$, since the bi-invariant 3-form $(X, Y, Z) \mapsto \langle [X, Y], Z\rangle$ is closed and not exact when G is not a torus; a torus that is a sphere is S^1. So a sphere S^n that is a Lie group has $n = 1$ or $n = 3$.

3.2. **The tautological bundle.** For $x \in \mathbb{K}$ let $L_x = \{(xc, c) : c \in \mathbb{K}\}$ and $L_\infty = \mathbb{K} \times \{0\}$, as in the Introduction. These are d-dimensional real subspaces of $\mathbb{K}^2$; L_x is the graph of ℓ_x over the second factor.

Proposition 3.3. *Every nonzero vector $(a, b) \in \mathbb{K}^2$ lies in exactly one of the subspaces L_x, $x \in \mathbb{K} \cup \{\infty\}$: in $L_{ab^{-1}}$ if $b \neq 0$, and in L_∞ if $b = 0$. Distinct subspaces meet only in 0.*

Proof. If $b = 0$ then $(a, 0) \in L_\infty$, and $(a, 0) \notin L_x$ for $x \in \mathbb{K}$, since $(a, 0) = (xc, c)$ forces $c = 0$ and then $a = 0$. If $b \neq 0$, then $(a, b) \notin L_\infty$, and $(a, b) = (xc, c)$ holds exactly when $c = b$ and $a = xb$. The equation $xb = a$ has the solution $x = ab^{-1}$, because $(ab^{-1})b = a$ by Lemma 3.1, and no other, because $xb = a$ implies $x = (xb)b^{-1} = ab^{-1}$. The last statement is a reformulation of the first. □

Identify $\mathbb{K} \cup \{\infty\}$ with $S^d \subset \mathbb{K} \times \mathbb{R}$ by the inverse stereographic projection from the north pole $N_0 = (0, 1)$,

$$s(x) = \frac{\big(2x,\ |x|^2 - 1\big)}{|x|^2 + 1} \quad (x \in \mathbb{K}), \qquad s(\infty) = N_0. \tag{3.3}$$

On $S^d \setminus \{s(0)\}$ we use the coordinate $y = x^{-1}$, with $y = 0$ at ∞. Since $x^{-1} = \bar{x}/|x|^2$,

$$s(y^{-1}) = \frac{\big(2\bar{y},\ 1 - |y|^2\big)}{1 + |y|^2}, \tag{3.4}$$

which is the inverse stereographic projection from the south pole composed with the reflection $y \mapsto \bar{y}$. Thus x and y are the two charts of the standard smooth structure on S^d, with transition map $y = x^{-1}$ on $\mathbb{K} \setminus 0$. For $y \in \mathbb{K}$ put $L'_y = \{(c, yc) : c \in \mathbb{K}\}$.

Lemma 3.4. *$L'_0 = L_\infty$, and $L'_{x^{-1}} = L_x$ for $x \neq 0$.*

Proof. The first equality is the definition. For $x \neq 0$ and $c \in \mathbb{K}$ put $c' = xc$; then $x^{-1}c' = x^{-1}(xc) = c$ by Lemma 3.1, so $(xc, c) = (c', x^{-1}c')$. Hence $L_x \subseteq L'_{x^{-1}}$, and both have dimension d. □

Let $L_\mathbb{K} = \{(p, v) \in S^d \times \mathbb{K}^2 : v \in L_p\}$, where L_p means L_x for $p = s(x)$.

Proof of Theorem A. The bundle. Over $U_0 = \mathbb{K}$ and $U_\infty = (\mathbb{K} \setminus 0) \cup \{\infty\}$ define

$$\varphi_0 \colon U_0 \times \mathbb{K} \to L_\mathbb{K}|_{U_0},\ (x, c) \mapsto \big(x, (xc, c)\big), \qquad \varphi_\infty \colon U_\infty \times \mathbb{K} \to L_\mathbb{K}|_{U_\infty},\ (y, c) \mapsto \big(y, (c, yc)\big),$$

the second written in the chart y. They are bijective by Lemma 3.4, with inverses $(x, (a, b)) \mapsto (x, b)$ and $(y, (a, b)) \mapsto (y, a)$, so they are homeomorphisms that are linear isomorphisms on fibres. On the overlap, $y = x^{-1}$ and

$$\varphi_\infty^{-1} \circ \varphi_0(x, c) = \big(x^{-1},\ xc\big), \tag{3.5}$$

so the transition function is $x \mapsto \ell_x \in \mathrm{GL}_d(\mathbb{R})$, which depends smoothly (indeed polynomially) on x. Hence $L_\mathbb{K}$ is a smooth vector bundle of rank d, and since $\varphi_0, \varphi_\infty$ are smooth maps into $S^d \times \mathbb{K}^2$ of constant rank, it is a smooth subbundle of the trivial bundle.

The sphere bundle and the Hopf map. Let $S(L_\mathbb{K})$ be the bundle of unit vectors, a closed subset of $S^d \times S^{2d-1}$. By Proposition 3.3 the projection $(p, v) \mapsto v$ is a bijection $S(L_\mathbb{K}) \to S^{2d-1}$. We compute the inverse. If $(a, b) = (xc, c) \in L_x$ is a unit vector, then $|c|^2(|x|^2 + 1) = 1$ by (D1), and $(xc)\bar{c} = |c|^2 x$ by Lemma 3.1, so

$$h_\mathbb{K}(xc, c) = \big(2(xc)\bar{c},\ |xc|^2 - |c|^2\big) = |c|^2\big(2x,\ |x|^2 - 1\big) = s(x),$$

and $h_\mathbb{K}(a, 0) = (0, |a|^2) = N_0 = s(\infty)$. So the inverse is $v \mapsto (h_\mathbb{K}(v), v)$, which is smooth, and $S(L_\mathbb{K}) \to S^{2d-1}$ is a diffeomorphism under which the bundle projection becomes $h_\mathbb{K}$. In particular $h_\mathbb{K}$ takes values in S^d; directly, $|h_\mathbb{K}(a, b)|^2 = 4|a|^2|b|^2 + (|a|^2 - |b|^2)^2 = (|a|^2 + |b|^2)^2$.

The fibres. The fibre of $h_{\mathbb{K}}$ over $s(x)$ is $L_x \cap S^{2d-1}$, the unit sphere of a d-dimensional linear subspace, that is, a great $(d-1)$-sphere.

Clutching. Let $D_0 = \{|x| \le 1\}$ and $D_\infty = \{|y| \le 1\}$ be the two closed hemispheres; they meet in the equator $|x| = |y| = 1$, where $y = x^{-1} = \bar{x}$. By (3.5) the bundle $L_{\mathbb{K}}$ is obtained from $D_0 \times \mathbb{K}$ and $D_\infty \times \mathbb{K}$ by identifying (x, c), $|x| = 1$, with $(\bar{x}, xc)$. This is the clutching construction with clutching function τ of (3.2). □

Corollary 3.5. *For $d = 1, 2, 4, 8$ the map $h_{\mathbb{K}}$ is a smooth fibre bundle $S^{d-1} \to S^{2d-1} \to S^d$, and hence a fibration. The fibre over $s(x)$ is $\{(xc, c) : |c|^2 = (1 + |x|^2)^{-1}\}$ for $x \in \mathbb{K}$ and $\{(a, 0) : |a| = 1\}$ over the north pole.*

Remark 3.6 (The four cases). For $\mathbb{K} = \mathbb{R}$ the bundle $L_{\mathbb{R}}$ is the Möbius line bundle over $S^1 = \mathbb{RP}^1$: the clutching function on $S^0 = \{\pm 1\}$ is $\tau(-1) = -1$, which reverses orientation, and $h_{\mathbb{R}}$ is the double covering $z \mapsto z^2$ of Example 2.7. For $\mathbb{K} = \mathbb{C}$ each L_x is a complex line, and $L_{\mathbb{C}}$ is the tautological line bundle over $\mathbb{CP}^1$. For $\mathbb{K} = \mathbb{H}$ each L_x is a right $\mathbb{H}$-submodule, since $(xc)q = x(cq)$, and $L_{\mathbb{H}}$ is the tautological right quaternionic line bundle over $\mathbb{HP}^1$. For $\mathbb{K} = \mathbb{O}$ the subspace L_x is not closed under right multiplication by octonions, and there is no module structure, but Theorem A gives a vector bundle all the same. This replaces the octonionic projective line, whose naive definition by proportional pairs is not an equivalence relation.[9]

Proposition 3.7 (Group actions)**.** *For $\mathbb{K} = \mathbb{C}$ the fibres of $h_{\mathbb{C}}$ are the orbits of the free action $(a, b) \cdot \lambda = (a\lambda, b\lambda)$ of $\mathrm{U}(1)$, and for $\mathbb{K} = \mathbb{H}$ they are the orbits of the free action $(a, b) \cdot q = (aq, bq)$ of $\mathrm{Sp}(1)$, so $h_{\mathbb{C}}$ and $h_{\mathbb{H}}$ are principal bundles. For $\mathbb{K} = \mathbb{O}$ the fibre through (a, b) is in general not $\{(ac, bc) : |c| = 1\}$.*

Proof. For $\mathbb{K}$ associative, $h(aq, bq) = (2aq\bar{q}\bar{b}, |a|^2 - |b|^2) = h(a, b)$ for $|q| = 1$, so each orbit lies in a fibre; it is a $(d-1)$-sphere, because the action is free, and therefore it is the whole fibre. For $\mathbb{O}$, the points (ac, bc) and (a, b) lie in the same fibre exactly when $(ac)(bc)^{-1} = ab^{-1}$, and this fails for generic a, b, c; Appendix C records an explicit triple. □

3.3. **The projective plane of the bundle.** For $d \ge 2$ let $D(L_{\mathbb{K}})$ be the unit disc bundle. It is a compact $2d$-manifold with boundary $S(L_{\mathbb{K}}) = S^{2d-1}$, and we put

$$\mathbb{KP}^2 = D(L_{\mathbb{K}}) \cup_{S^{2d-1}} D^{2d}, \tag{3.6}$$

glued along the identity of $S^{2d-1} = \partial D^{2d}$. For $\mathbb{K} = \mathbb{C}, \mathbb{H}$ this is the complex or quaternionic projective plane, and for $\mathbb{K} = \mathbb{O}$ it is the Cayley plane;[10] we keep the notation $\mathbb{KP}^2$ in all three cases.

Proposition 3.8. *Let $d \in \{2, 4, 8\}$.*

(i) *$\mathbb{KP}^2$ is a closed, simply connected, hence orientable, topological $2d$-manifold.*
(ii) *$\mathbb{KP}^2$ is homeomorphic to the mapping cone $C_h = S^d \cup_{h_{\mathbb{K}}} D^{2d}$ of $h_{\mathbb{K}}$.*
(iii) *$H^j(\mathbb{KP}^2; \mathbb{Z}) \cong \mathbb{Z}$ for $j = 0, d, 2d$ and $H^j(\mathbb{KP}^2; \mathbb{Z}) = 0$ otherwise.*

Proof. (ii) The disc bundle $D(L_{\mathbb{K}})$ is homeomorphic to the mapping cylinder of the projection $S(L_{\mathbb{K}}) \to S^d$: the map $(v, t) \mapsto tv$ from $S(L_{\mathbb{K}}) \times [0, 1]$ onto $D(L_{\mathbb{K}})$ identifies exactly the points with $t = 0$ in the same fibre. Gluing D^{2d} to the mapping cylinder along its free end $S(L_{\mathbb{K}}) = S^{2d-1}$ gives the mapping cone. (i) A manifold with boundary glued to another along a homeomorphism of their boundaries is a manifold, since boundaries have collar neighbourhoods. The space is compact and has no boundary. By (ii) it is a CW complex with one cell in each of the dimensions

[9] The relation $(a, b) \sim (ac, bc)$ for $c \neq 0$ is not transitive on $\mathbb{O}^2$, because $(ac)c' \neq a(cc')$ in general. Proposition 3.3 replaces it by the relation $ab^{-1} = a'b'^{-1}$.

[10] We only use the topological description (3.6), which makes no reference to octonionic lines in $\mathbb{O}^3$.

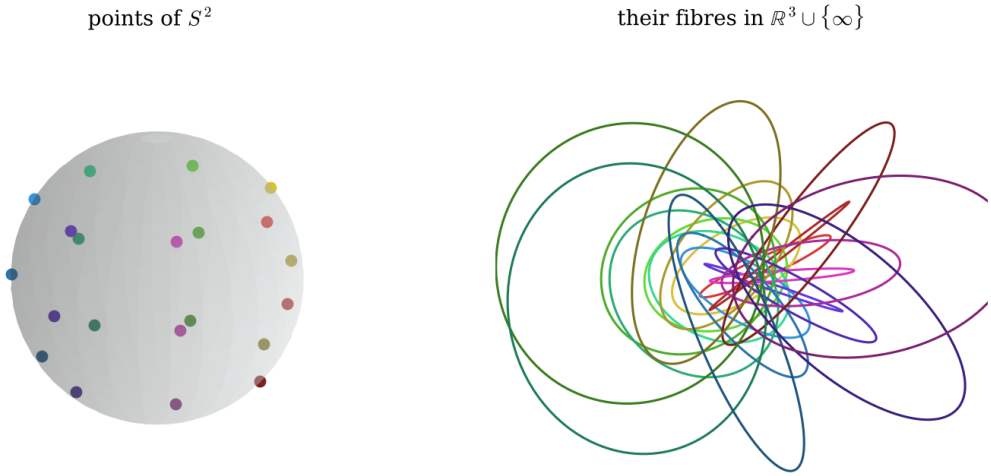


FIGURE 1. Left: points on the base S^2. Right: the corresponding fibres of $h\colon S^3 \to S^2$ after stereographic projection to $\mathbb{R}^3$, in the same colours. Points on one circle of latitude give fibres on one torus; the north pole gives the circle at infinity (here the vertical axis together with ∞), and the south pole gives the unit circle in the horizontal plane.

0, d and $2d$; as $d \geq 2$ there are no 1-cells, so it is simply connected, and a simply connected manifold is orientable. (iii) The cellular cochain complex has one generator in each of the degrees $0, d, 2d$ and all differentials zero, because no two cells have adjacent dimensions. □

4. The Hopf invariant

This section proves Theorem B. Section 4.1 defines the invariant and proves the rules for composition that are used throughout; Section 4.3 computes the degree of the octonionic power maps.

4.1. **Definition and composition rules.** Let $n \geq 2$ and $f\colon S^{2n-1} \to S^n$. The mapping cone $C_f = S^n \cup_f D^{2n}$ is a CW complex with cells in dimensions 0, n and $2n$, so

$$H^n(C_f;\mathbb{Z}) \cong \mathbb{Z}\,\alpha_f, \qquad H^{2n}(C_f;\mathbb{Z}) \cong \mathbb{Z}\,\beta_f,$$

where α_f restricts to the chosen generator of $H^n(S^n)$ and β_f corresponds to the chosen generator of $H^{2n}(D^{2n}, S^{2n-1})$ under the isomorphisms $H^{2n}(C_f) \cong H^{2n}(C_f, S^n) \cong H^{2n}(D^{2n}, S^{2n-1})$.

Definition 4.1. The *Hopf invariant* $\mathrm{H}(f) \in \mathbb{Z}$ is defined by $\alpha_f \smile \alpha_f = \mathrm{H}(f)\,\beta_f$.

If $f \simeq f'$, then C_f and $C_{f'}$ are homotopy equivalent by a map that is the identity on S^n and of degree one on the top cell, so $\mathrm{H}(f)$ depends only on the homotopy class of f. Changing the orientation of S^{2n-1} changes the sign of H, and changing that of S^n does not. The function $\mathrm{H}\colon \pi_{2n-1}(S^n) \to \mathbb{Z}$ is a homomorphism [Hat02, Proposition 4B.1]; we use this only in Corollary 4.11 and in Section 9.

Proposition 4.2. *If n is odd then $\mathrm{H}(f) = 0$ for every $f\colon S^{2n-1} \to S^n$.*

Proof. By graded commutativity $\alpha_f^2 = (-1)^{n^2}\alpha_f^2 = -\alpha_f^2$, so $2\mathrm{H}(f)\beta_f = 0$ in $H^{2n}(C_f) \cong \mathbb{Z}$. □

Proposition 4.3 (Composition)**.** *Let $f\colon S^{2n-1} \to S^n$, let $g\colon S^{2n-1} \to S^{2n-1}$ have degree m and let $\varphi\colon S^n \to S^n$ have degree k. Then*

$$\mathrm{H}(f \circ g) = m\,\mathrm{H}(f), \qquad \mathrm{H}(\varphi \circ f) = k^2\,\mathrm{H}(f).$$

Proof. Let $\hat{g}\colon D^{2n} \to D^{2n}$, $\hat{g}(tu) = t\,g(u)$, be the cone on g. It maps S^{2n-1} to itself by g, and as a map of pairs $(D^{2n}, S^{2n-1}) \to (D^{2n}, S^{2n-1})$ it has degree m, by the five lemma applied to the long exact sequences of the pairs and the boundary isomorphism $H^{2n}(D^{2n}, S^{2n-1}) \cong H^{2n-1}(S^{2n-1})$. The identity of S^n together with $\hat{g}$ defines a map $\Phi\colon C_{f\circ g} \to C_f$, since a point $u \in \partial D^{2n}$ is attached at $f(g(u))$ in $C_{f\circ g}$ and $\hat{g}(u) = g(u)$ is attached at $f(g(u))$ in C_f:

$$\begin{array}{ccccc} S^{2n-1} & \xrightarrow{\ g\ } & S^{2n-1} & \xrightarrow{\ f\ } & S^n \\ \downarrow & & \downarrow & & \\ D^{2n} & \xrightarrow{\ \hat{g}\ } & D^{2n} & & \end{array} \qquad \rightsquigarrow \qquad \Phi\colon C_{f\circ g} = S^n \cup_{fg} D^{2n} \longrightarrow S^n \cup_f D^{2n} = C_f.$$

Then $\Phi^*\alpha_f = \alpha_{fg}$ and $\Phi^*\beta_f = m\beta_{fg}$, so applying Φ^* to $\alpha_f^2 = \mathrm{H}(f)\beta_f$ gives $\alpha_{fg}^2 = m\mathrm{H}(f)\beta_{fg}$. For the second identity let $\Psi\colon C_f \to C_{\varphi f}$ be φ on S^n and the identity on D^{2n}. Then $\Psi^*\alpha_{\varphi f} = k\alpha_f$ and $\Psi^*\beta_{\varphi f} = \beta_f$, and applying Ψ^* to $\alpha_{\varphi f}^2 = \mathrm{H}(\varphi f)\beta_{\varphi f}$ gives $k^2\mathrm{H}(f)\beta_f = \mathrm{H}(\varphi f)\beta_f$. □

Proposition 4.4. *If n is even, then $\mathrm{H}([\iota_n, \iota_n]) = \pm 2$.*

Proof. Write $S^n \times S^n = (S^n \vee S^n) \cup_w D^{2n}$ as in Section 2, and let $q\colon S^n \times S^n \to C = C_{\nabla\circ w}$ be the fold map ∇ on $S^n \vee S^n$ and the identity on the top cell; it is well defined for the same reason as Φ above. Let $a, b \in H^n(S^n \times S^n)$ be the pull-backs of the generator along the two projections. Restriction to the n-skeleton is an isomorphism on H^n for both spaces, and ∇^* sends the generator to $a + b$; hence $q^*\alpha = a + b$, where $\alpha = \alpha_{\nabla w}$. On the top cell q is the identity, so $q^*\beta = \pm ab$, since ab generates $H^{2n}(S^n \times S^n)$ by the Künneth theorem. Now $a^2 = b^2 = 0$, as they are pulled back from $H^{2n}(S^n) = 0$, and $ba = (-1)^{n^2}ab = ab$ since n is even, so

$$q^*(\alpha^2) = (a + b)^2 = ab + ba = 2ab = \pm 2\,q^*\beta.$$

As q^* is an isomorphism on H^{2n}, $\alpha^2 = \pm 2\beta$. □

4.2. **Hopf invariant one.**

Proof of Theorem B(i). By Proposition 3.8(ii) the mapping cone of $h_{\mathbb{K}}$ is homeomorphic to $\mathbb{KP}^2$, a closed simply connected $2d$-manifold, by a homeomorphism that is the identity on S^d and of degree ± 1 on the top cell. Its cohomology is free, and by Poincaré duality (input (I2), in the form [Hat02, Proposition 3.38]) the cup product pairing $H^d(\mathbb{KP}^2) \times H^d(\mathbb{KP}^2) \to H^{2d}(\mathbb{KP}^2) \cong \mathbb{Z}$ is nonsingular. Since H^d is generated by α, this says $\alpha^2 = \pm\beta$. □

Remark 4.5. The proof uses that C_h is a manifold, and so it applies to any fibre bundle $p\colon S^{2d-1} \to S^d$ with fibre S^{d-1}, $d \geq 2$: coning the fibres turns p into a bundle with fibre D^d, which is a manifold with boundary S^{2d-1}, and gluing D^{2d} gives a closed manifold homeomorphic to C_p. With Adams' theorem (I4) this gives the following statement, quoted in the Introduction.

Corollary 4.6. *If $S^{d-1} \to S^{2d-1} \xrightarrow{p} S^d$ is a fibre bundle with $d \geq 2$, then $\mathrm{H}(p) = \pm 1$ and $d \in \{2, 4, 8\}$.*

4.3. **Power maps.** By Artin's theorem the powers x^m of an octonion are well defined, since x generates an associative subalgebra. For $d \in \{2, 4, 8\}$ let

$$p_m\colon S^{d-1} \to S^{d-1}, \qquad p_m(x) = x^m,$$

which takes values in the unit sphere by (D1). Every $x \in S^{d-1} \setminus \{\pm 1\}$ can be written uniquely as $x = \cos\theta + u \sin\theta$ with $0 < \theta < \pi$ and u a unit vector orthogonal to 1; such a u satisfies $u^2 = -1$ by (D2), so $\mathbb{C}_u = \mathbb{R} \oplus \mathbb{R}u$ is a subalgebra isomorphic to $\mathbb{C}$, and $x^m = \cos m\theta + u \sin m\theta$.

Lemma 4.7. *Let $x = \cos\theta + u\sin\theta$ as above and let $W_u \subset \mathbb{K}$ be the orthogonal complement of $\mathbb{C}_u$. Then W_u is tangent to S^{d-1} at x and at x^m, and*

$$Dp_m(x)(w) = \frac{\sin m\theta}{\sin\theta}\, w \qquad (w \in W_u), \qquad Dp_m(x)(ux) = m\, u\, x^m.$$

Proof. The second formula is the derivative of $e^{i\theta} \mapsto e^{im\theta}$ in $\mathbb{C}_u$, since $ux = \frac{d}{d\theta}x$. For the first, x and w lie in a subalgebra that is associative by Artin's theorem, so $Dp_m(x)(w) = \sum_{j=0}^{m-1} x^j w x^{m-1-j}$, computed by expanding $(x+tw)^m$. Since $w \perp 1$ and $w \perp u$, (D3) gives $\mathrm{Re}(w) = 0$ and $\mathrm{Re}(u\bar{w}) = 0$, hence $\bar{w} = -w$ and $uw = -wu$ (the latter because $uw + wu = -(u\bar{w} + w\bar{u}) = -2\langle u, w\rangle = 0$). Therefore $wx = \bar{x}w$, and $x^j w x^{m-1-j} = x^j \bar{x}^{m-1-j} w = x^{2j-m+1} w$ because $\bar{x} = x^{-1}$ in $\mathbb{C}_u$. Finally

$$\sum_{j=0}^{m-1} x^{2j-m+1} = \sum_{j=0}^{m-1} \big(\cos(2j-m+1)\theta + u\sin(2j-m+1)\theta\big) = \frac{\sin m\theta}{\sin\theta},$$

the sine terms cancelling in pairs $j \leftrightarrow m-1-j$. The vectors $w \in W_u$ are orthogonal to x and to x^m, both of which lie in $\mathbb{C}_u$, so they are tangent to the sphere at both points. □

Proposition 4.8. *For $d \in \{2, 4, 8\}$ and $m \geq 1$ the power map p_m has degree m.*

Proof. Let $y = \cos\varphi + v\sin\varphi$ with $0 < \varphi < \pi$, a non-real point of S^{d-1}. If $x^m = y$, then $y \in \mathbb{C}_x$, and $\mathbb{C}_x = \mathbb{C}_y = \mathbb{R} \oplus \mathbb{R}y$ because a two-dimensional subalgebra containing the non-real element y is spanned by 1 and y. So the solutions of $x^m = y$ are the m roots of y in the circle $\mathbb{C}_y \cap S^{d-1}$, and none of them is real. Write such a root as $x = \cos\theta + v\sin\theta$ with $\theta \in (-\pi, \pi) \setminus \{0\}$; the formulas of Lemma 4.7 are unchanged under $(u, \theta) \mapsto (-u, -\theta)$, so they hold with $u = v$. We have $\sin\theta \neq 0$ and $\sin m\theta = \pm\sin\varphi \neq 0$. By Lemma 4.7, in the bases $(vx, w_1, \dots, w_{d-2})$ of $T_x S^{d-1}$ and $(vy, w_1, \dots, w_{d-2})$ of $T_y S^{d-1}$, with (w_j) a fixed basis of W_v, the derivative is $\mathrm{diag}(m, c, \dots, c)$ with $c = \sin m\theta / \sin\theta \neq 0$. Its determinant $m\, c^{d-2}$ is positive, since $d-2 \in \{0, 2, 6\}$ is even. Both bases are positively oriented or both negatively: the frames $(x, vx, w_1, \dots)$ and $(y, vy, w_1, \dots)$ of $\mathbb{R}^d$ differ by a rotation of the plane $\mathbb{C}_v$ that fixes W_v. So y is a regular value with m preimages, each counted with sign $+1$. □

Remark 4.9. The scalar $c = \sin m\theta / \sin\theta$ vanishes at $\theta = k\pi/m$, $0 < k < m$, where $x^m = \pm 1$; the critical points of p_m for $d \geq 4$ form the $(d-2)$-spheres $\{\cos(k\pi/m) + u\sin(k\pi/m) : u \perp 1,\ |u| = 1\}$, which p_m collapses to the points $(-1)^k$. For $d = 2$ there are no critical points.

Proposition 4.10. *For $d \in \{2, 4, 8\}$ and $m \geq 1$ the map $g_m \colon S^{2d-1} \to S^{2d-1}$, $g_m(a, b) = (a^m, b)/|(a^m, b)|$, has degree m, and it fixes the point $(1, 0)$. The map $\kappa(a, b) = (\bar{a}, b)$ has degree -1.*

Proof. Write $a = \rho x$ with $\rho = |a|$ and $|x| = 1$. For $t \in [0, 1]$ the maps

$$F_t(\rho x, b) = \big(\rho^{(1-t)m+t} x^m,\ b\big)$$

are continuous on $\mathbb{K}^2$ (the exponent is at least 1) and vanish only at 0, so $F_t/|F_t|$ is a homotopy of self-maps of S^{2d-1} fixing $(1, 0)$, from g_m to $g'(a, b) = (|a|\, p_m(a/|a|),\ b)$, extended by $g'(0, b) = (0, b)$; note that $(|a|\, p_m(a/|a|), b)$ already has norm one. On the open dense subset $\{a \neq 0,\ b \neq 0\}$ of S^{2d-1} we use the diffeomorphism

$$(0, \tfrac{\pi}{2}) \times S^{d-1} \times S^{d-1} \to \{a \neq 0, b \neq 0\}, \qquad (s, x, z) \mapsto (x\cos s,\ z\sin s),$$

in which g' becomes $(s,x,z)\mapsto(s,p_m(x),z)$. A regular value y of p_m gives a regular value $(y\cos s, z\sin s)$ of g', whose preimages $(x\cos s, z\sin s)$, $x^m=y$, are m points at which the Jacobian in these coordinates has the sign of $\det Dp_m(x)$, that is $+1$. The same chart is used in source and target, so the sign relative to the orientation of S^{2d-1} is also $+1$, and $\deg g_m=\deg g'=m$. The map κ is the restriction of a linear map of $\mathbb{R}^{2d}$ that negates the $d-1$ imaginary coordinates of a, so its degree is $(-1)^{d-1}=-1$. □

Proof of Theorem B(ii) *and* (iii). Part (ii) is Propositions 4.8 and 4.10. Since g_m is a based map of degree m, Lemma 2.4 gives $[h_{\mathbb{K}}\circ g_m]=m[h_{\mathbb{K}}]$ in $\pi_{2d-1}(S^d)$, and Proposition 4.3 gives $\mathrm{H}(h_{\mathbb{K}}\circ g_m)=m\mathrm{H}(h_{\mathbb{K}})=\pm m$ and $\mathrm{H}(h_{\mathbb{K}}\circ g_m\circ\kappa)=\mp m$. Because $h_{\mathbb{K}}(\lambda v)=\lambda^2 h_{\mathbb{K}}(v)$ for real λ,

$$h_{\mathbb{K}}\circ g_m(a,b)=\frac{\bigl(2a^m\bar{b},\ |a|^{2m}-|b|^2\bigr)}{|a|^{2m}+|b|^2}, \tag{4.1}$$

whose numerator and denominator are polynomials in the real coordinates of (a,b). With the constant map for 0, every integer is the Hopf invariant of such a map. □

Corollary 4.11. *The image of* $\mathrm{H}\colon\pi_{2n-1}(S^n)\to\mathbb{Z}$ *is* 0 *for* n *odd,* $\mathbb{Z}$ *for* $n=2,4,8$*, and* $2\mathbb{Z}$ *for every other even* n*.*

Proof. The first case is Proposition 4.2. For n even the image is a subgroup containing 2 by Proposition 4.4, so it is $\mathbb{Z}$ or $2\mathbb{Z}$; it is $\mathbb{Z}$ for $n=2,4,8$ by Theorem B(i), and $2\mathbb{Z}$ otherwise by Adams' theorem (I4). □

Remark 4.12. For $n=2$ the group $\pi_3(S^2)\cong\mathbb{Z}$ (Theorem C) is detected by H, so Proposition 4.4 gives $[\iota_2,\iota_2]=\pm2[h]$. For $n=4$ it gives $[\iota_4,\iota_4]\equiv\pm2[h_{\mathbb{H}}]$ modulo the torsion subgroup $\mathbb{Z}/12$ of $\pi_7(S^4)$; the exact formula $[\iota_4,\iota_4]=2\nu_4-E\nu'$ in Toda's notation is [Tod62, Chapter V]. Proposition 4.3 also shows that $\mathrm{H}(h\circ\varphi)=k$ for a self-map φ of S^3 of degree k, while $\mathrm{H}(\varphi'\circ h)=k^2$ for a self-map φ' of S^2 of degree k; both appear again in Section 6, as $[z^k:w]$ and $[z^k:w^k]$.

5. Homotopy groups from the Hopf bundles

5.1. **The splitting.**

Proof of Theorem C. Let $d\in\{2,4,8\}$ and let $F=h_{\mathbb{K}}^{-1}(N_0)=\{(a,0):|a|=1\}\cong S^{d-1}$ be the fibre over the north pole. By Corollary 3.5, $h_{\mathbb{K}}$ is a fibration. The inclusion $j\colon F\to S^{2d-1}$ is null-homotopic, because $\pi_{d-1}(S^{2d-1})=0$ by Proposition 2.2; explicitly,

$$G\colon F\times[0,\tfrac{\pi}{2}]\to S^{2d-1},\qquad G(a,s)=(a\cos s,\ \sin s) \tag{5.1}$$

contracts F to the point $(0,1)$. All three spaces are path connected, and $\pi_{i-1}(S^{d-1})$ is abelian for $i\geq2$ (it is $\pi_1(S^1)\cong\mathbb{Z}$ or zero when $i=2$). Lemma 2.8 gives the split exact sequence

$$0\to\pi_i(S^{2d-1})\xrightarrow{(h_{\mathbb{K}})_*}\pi_i(S^d)\xrightarrow{\partial}\pi_{i-1}(S^{d-1})\to0.$$

For $d=2$ and $i\geq3$ the third group is $\pi_{i-1}(S^1)=0$ by Example 2.6, so $(h_{\mathbb{C}})_*$ is an isomorphism. The values $\pi_6(S^3)\cong\mathbb{Z}/12$, $\pi_7(S^3)\cong\mathbb{Z}/2$ and $\pi_{14}(S^7)\cong\mathbb{Z}/120$ are taken from Toda's tables (I6), [Tod62, Chapter XIV], and give $\pi_7(S^4)\cong\mathbb{Z}\oplus\mathbb{Z}/12$, $\pi_8(S^4)\cong(\mathbb{Z}/2)^2$ and $\pi_{15}(S^8)\cong\mathbb{Z}\oplus\mathbb{Z}/120$. The infinite cyclic summand of $\pi_{2d-1}(S^d)$ is $(h_{\mathbb{K}})_*\pi_{2d-1}(S^{2d-1})$, generated by $[h_{\mathbb{K}}]$, and by Proposition 4.3 the Hopf invariant of $(h_{\mathbb{K}})_*[g]=[h_{\mathbb{K}}\circ g]$ is $\pm\deg g$; so it detects this summand. □

The splitting map of Lemma 2.8 can be identified.

Proposition 5.1. *For the null-homotopy* (5.1)*, the splitting* $\sigma\colon \pi_{i-1}(S^{d-1}) \to \pi_i(S^d)$ *of* (2.2) *is the suspension homomorphism* E*, up to a sign that depends only on orientation conventions. Hence*

$$\pi_i(S^d) = (h_{\mathbb{K}})_*\pi_i(S^{2d-1}) \;\oplus\; E\,\pi_{i-1}(S^{d-1}) \qquad (d = 2, 4, 8,\ i \geq 2),$$

and the suspensions $E\colon \pi_{i-1}(S^3) \to \pi_i(S^4)$ *and* $E\colon \pi_{i-1}(S^7) \to \pi_i(S^8)$ *are injective for every* i*.*

Proof. Composing G with $h_{\mathbb{K}}$ gives

$$h_{\mathbb{K}}\big(G(a,s)\big) = \big(2a\cos s \sin s,\ \cos^2 s - \sin^2 s\big) = \big(a \sin 2s,\ \cos 2s\big),$$

which, as s runs from 0 to $\pi/2$, moves the point a of the fibre along the meridian of S^d from the north pole through the direction a to the south pole. For $\beta\colon S^{i-2} \to F$, the element $\sigma[\beta]$ is represented by $(u,s) \mapsto (\beta(u)\sin 2s, \cos 2s)$ on $S^{i-2} \times [0, \pi/2]$, which collapses $S^{i-2} \times \{0\}$ to the north pole and $S^{i-2} \times \{\pi/2\}$ to the south pole. This is the unreduced suspension of β followed by the identification $\Sigma S^{d-1} \cong S^d$ given by meridians; as an element of $\pi_i(S^d)$ it is $\pm E[\beta]$, the unreduced and reduced suspensions being homotopy equivalent. The last statement holds because $\sigma = \pm E$ is a split injection. □

Remark 5.2. For $d = 2$ and $i = 2$ the proposition gives $\pi_2(S^2) = E\pi_1(S^1) \cong \mathbb{Z}$, generated by $E[\mathrm{id}_{S^1}] = [\mathrm{id}_{S^2}]$, in accordance with Proposition 2.3. For $d = 4$ it recovers the classical decomposition $\pi_i(S^4) \cong \pi_i(S^7) \oplus \pi_{i-1}(S^3)$ with the second summand embedded by suspension [Tod62, Chapter V].

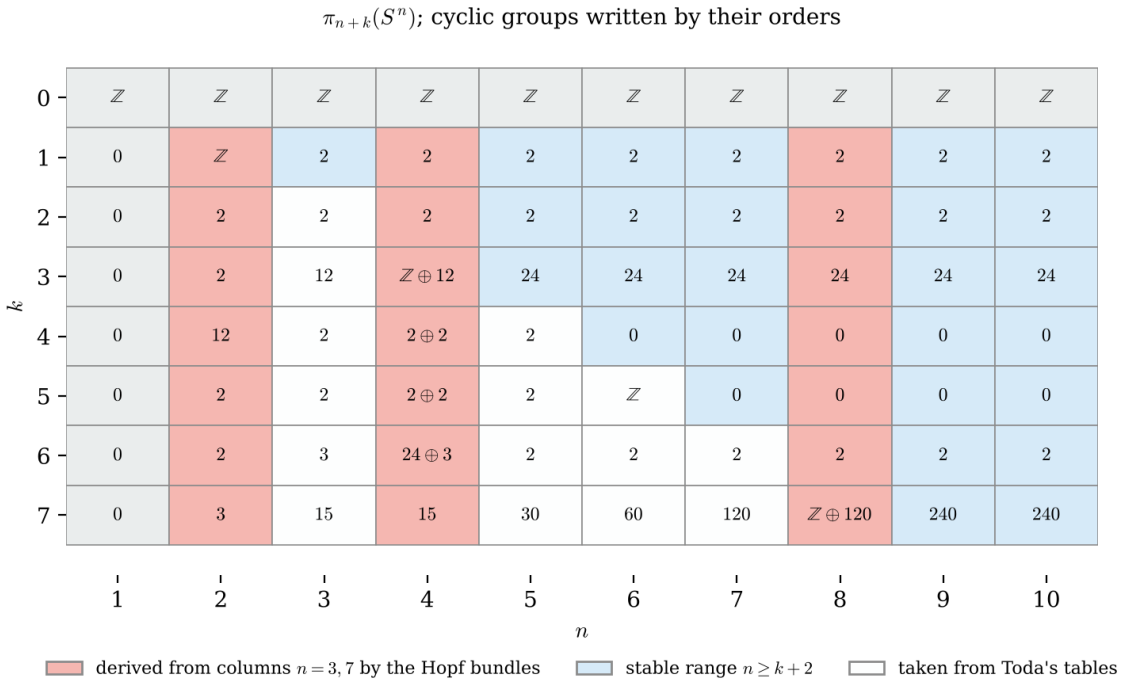
5.2. **The table of unstable groups.** Table 1 lists $\pi_{n+k}(S^n)$ for $k \leq 7$ and $n \leq 9$, with the stable value in the last column. All entries are taken from Toda [Tod62], but the entries in the columns $n = 2, 4, 8$ are also consequences of Theorem C and the entries in the columns $n = 3$, $n = 7$; the columns $n \geq k + 2$ are constant by Freudenthal's theorem. Figure 2 shows the same table with the three kinds of entries distinguished. That the table is consistent with Theorem C and with Theorem 2.10, as abelian groups, is checked by the program of Appendix C.

$k\backslash n$	1	2	3	4	5	6	7	8	9	π_k^s
0	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$
1	0	$\mathbb{Z}$	2	2	2	2	2	2	2	2
2	0	2	2	2	2	2	2	2	2	2
3	0	2	12	$\mathbb{Z}\oplus 12$	24	24	24	24	24	24
4	0	12	2	2^2	2	0	0	0	0	0
5	0	2	2	2^2	2	$\mathbb{Z}$	0	0	0	0
6	0	2	3	$24\oplus 3$	2	2	2	2	2	2
7	0	3	15	15	30	60	120	$\mathbb{Z}\oplus 120$	240	240

TABLE 1. The groups $\pi_{n+k}(S^n)$ for $0 \leq k \leq 7$, after Toda [Tod62]; an integer m stands for $\mathbb{Z}/m$ and 2^2 for $(\mathbb{Z}/2)^2$. The column $n = 2$ equals the column $n = 3$ shifted down by one row; the column $n = 4$ is the sum of the column $n = 7$ shifted down by three rows and the column $n = 3$; the column $n = 8$ is the sum of the column $n = 7$ and the group $\pi_{8+k}(S^{15})$, which is $\mathbb{Z}$ for $k = 7$ and 0 for $k < 7$.

Example 5.3. Reading the table through Theorem C:

(a) $\pi_4(S^2) \cong \pi_4(S^3) \cong \mathbb{Z}/2$, $\pi_5(S^2) \cong \pi_5(S^3) \cong \mathbb{Z}/2$, $\pi_6(S^2) \cong \pi_6(S^3) \cong \mathbb{Z}/12$.
(b) $\pi_8(S^4) \cong \pi_8(S^7) \oplus \pi_7(S^3) \cong (\mathbb{Z}/2)^2$ and $\pi_9(S^4) \cong \pi_9(S^7) \oplus \pi_8(S^3) \cong (\mathbb{Z}/2)^2$.
(c) $\pi_{10}(S^4) \cong \pi_{10}(S^7) \oplus \pi_9(S^3) \cong \mathbb{Z}/24 \oplus \mathbb{Z}/3$.
(d) $\pi_i(S^8) \cong \pi_{i-1}(S^7)$ for $8 \leq i \leq 14$, and $\pi_{15}(S^8) \cong \mathbb{Z} \oplus \mathbb{Z}/120$.



k \ n	1	2	3	4	5	6	7	8	9	10
0	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$
1	0	$\mathbb{Z}$	2	2	2	2	2	2	2	2
2	0	2	2	2	2	2	2	2	2	2
3	0	2	12	$\mathbb{Z}\oplus 12$	24	24	24	24	24	24
4	0	12	2	$2\oplus 2$	2	0	0	0	0	0
5	0	2	2	$2\oplus 2$	2	$\mathbb{Z}$	0	0	0	0
6	0	2	3	$24\oplus 3$	2	2	2	2	2	2
7	0	3	15	15	30	60	120	$\mathbb{Z}\oplus 120$	240	240

FIGURE 2. The table of $\pi_{n+k}(S^n)$. Dark cells are derived from the columns $n = 3$ and $n = 7$ by the Hopf bundles (Theorem C); light cells lie in the stable range $n \geq k + 2$ (Theorem 2.10); white cells are taken from Toda's tables.

In each case the summand $\pi_i(S^{2d-1})$ is the image of composition with the Hopf map, and the summand $\pi_{i-1}(S^{d-1})$ is the image of the suspension.

5.3. **Generators.** In Toda's notation [Tod62] the Hopf maps are $\eta_2 = [h_{\mathbb{C}}]$, $\nu_4 = [h_{\mathbb{H}}]$ and $\sigma_8 = [h_{\mathbb{O}}]$, and η_n, ν_n, σ_n denote their iterated suspensions. The stable stems in degrees 1, 3 and 7 are cyclic of orders 2, 24 and 240, generated by η, ν and σ [Tod62, Chapters V and XIV]. Before suspension the Hopf maps have infinite order, since their Hopf invariant is ± 1 (Theorem B); after suspension they have finite order and generate the image of the J-homomorphism in these degrees (Section 9). For instance, $E^2\nu' = 2\nu_5$ for the generator ν' of $\pi_6(S^3) \cong \mathbb{Z}/12$ [Tod62, Chapter V], so the summand $E\pi_6(S^3)$ of $\pi_7(S^4)$ maps onto the subgroup of index two of $\pi_8(S^5) \cong \mathbb{Z}/24$, and the summand $\mathbb{Z}\nu_4$ maps onto the whole group.

6. Linking and holomorphic maps

6.1. **Linking numbers.** Let $K_1, K_2 \subset S^3$ be disjoint oriented closed 1-manifolds. Since $H_1(S^3) = 0$, K_1 bounds an oriented surface Σ (a Seifert surface, or more generally a 2-chain), which may be chosen transverse to K_2, and the *linking number* $\mathrm{lk}(K_1, K_2)$ is the algebraic intersection number $\Sigma \cdot K_2$. It does not depend on Σ, it is symmetric, and for $K_1, K_2 \subset \mathbb{R}^3$ it is given by the Gauss integral

$$\mathrm{lk}(K_1, K_2) = \frac{1}{4\pi} \oint_{K_1} \oint_{K_2} \frac{(\mathbf{x} - \mathbf{y}) \cdot (d\mathbf{x} \times d\mathbf{y})}{|\mathbf{x} - \mathbf{y}|^3}. \tag{6.1}$$

The integral (6.1) is used in Appendix C to check the computations of this section numerically.

Orient the fibres of $h\colon S^3 \to S^2$ by the circle action $\lambda \mapsto (\lambda z, \lambda w)$.

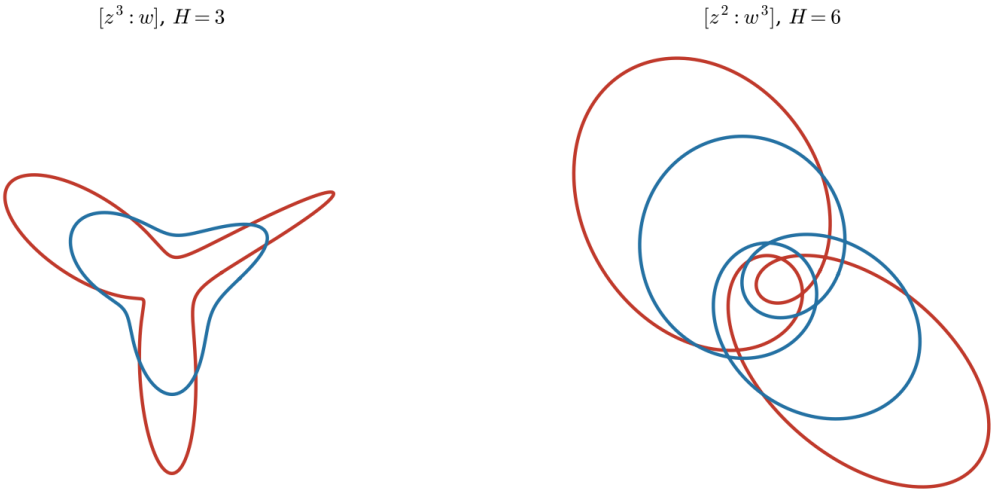


FIGURE 3. Two fibres of $[z^3 : w]$, with Hopf invariant 3 (left), and of $[z^2 : w^3]$, with Hopf invariant 6 (right), after stereographic projection. Each fibre lies on a torus about the axis and is a torus link of type (p, q); the Gauss integral (6.1) gives linking numbers 3 and 6 (Appendix C).

Proposition 6.1. *Any two distinct fibres of h have linking number ε, where $\varepsilon \in \{\pm 1\}$ depends only on the orientation of S^3.*

Proof. The fibre over the north pole N_0 is $C_\infty = \{(z, 0) : |z| = 1\}$. It bounds the hemisphere

$$\Sigma = \{(z, t) \in S^3 : t \in \mathbb{R},\ t \geq 0\},$$

a 2-disc. By Corollary 3.5 the fibre over $s(x)$, $x \in \mathbb{C}$, is $C_x = \{(xc, c) : |c| = (1 + |x|^2)^{-1/2}\}$, and it meets Σ exactly where c is real and positive, at one point. The intersection is transverse: along C_x, written as $c = \rho e^{i\phi}$, the function $\operatorname{Im} w = \rho \sin \phi$ that cuts out Σ near the intersection has nonzero derivative ρ at $\phi = 0$. So $\operatorname{lk}(C_\infty, C_x) = \pm 1$, and the sign is constant in x because $\mathbb{C}$ is connected and the fibres C_x vary continuously. Now let C_p, C_q be any two distinct fibres. The group $\mathrm{U}(2)$ acts on S^3 by orientation-preserving isometries commuting with the circle action, so it permutes the oriented fibres, and it acts transitively on S^2 through $\mathrm{U}(2) \to \mathrm{SO}(3)$. Choosing $g \in \mathrm{U}(2)$ with $gC_p = C_\infty$ gives $\operatorname{lk}(C_p, C_q) = \operatorname{lk}(C_\infty, gC_q)$, which is the constant found above. □

The connection with the Hopf invariant is Hopf's original definition [Hop31]; see [BT82, Section 17] for a proof.

Theorem 6.2 (Linking formula)**.** *Let $f\colon S^3 \to S^2$ be smooth and let $p \neq q$ be regular values. Orient $f^{-1}(p)$ and $f^{-1}(q)$ as preimages. Then $\mathrm{H}(f) = \pm \operatorname{lk}(f^{-1}(p), f^{-1}(q))$, with a universal sign.*

We fix the orientation of S^3 (the boundary orientation of the unit ball in $\mathbb{C}^2$ with its complex orientation) and choose the generators in Definition 4.1 so that $\mathrm{H}(h) = 1$. With these choices the universal sign is ε, by Proposition 6.1 applied to $f = h$: the fibres of h, oriented by the circle action, are its preimages with their preimage orientation up to one fixed sign.

6.2. **Holomorphic maps.** Let $P, Q \in \mathbb{C}[z,w]$ have no common zero on S^3, put $F = (P,Q)\colon \mathbb{C}^2 \to \mathbb{C}^2$ and

$$[P:Q] = h \circ \frac{F}{|F|}\colon S^3 \to S^2,$$

which is the map $u \mapsto [P(u):Q(u)] \in \mathbb{CP}^1 = S^2$ in the chart (3.3) with $x = P/Q$; the formula of the Introduction follows from $h(\lambda v) = \lambda^2 h(v)$ for real λ. Let $Z = \{u \in \mathbb{C}^2 : P(u) = Q(u) = 0\}$.

Lemma 6.3. *Let $C \subset \mathbb{C}^2$ be an irreducible affine plane curve that does not meet S^3. Then C does not meet the closed ball $\bar{B}^4$.*

Proof. The closure of C in $\mathbb{CP}^2$ is an irreducible projective curve, and C is its complement of the finitely many points at infinity, so C is connected in the classical topology [Mum76, Corollary (4.16)]. As $C \cap S^3 = \emptyset$, C is the disjoint union of the open sets $C \cap B^4$ and $C \setminus \bar{B}^4$, so if C meets $\bar{B}^4$ then $C \subset B^4$ is bounded. But no affine curve $C = \{D = 0\}$, D irreducible, is bounded. If D has degree $k \geq 1$ in w, with leading coefficient $a(z)$, then for every z_1 with $a(z_1) \neq 0$ the polynomial $D(z_1, \cdot)$ has a root, so the projection of C to the z-coordinate contains the complement of a finite set. If D has degree 0 in w, it is an irreducible polynomial in z alone, that is $D = z - z_0$ up to a scalar, and $C = \{z_0\} \times \mathbb{C}$. □

Lemma 6.4 (Finiteness)**.** *Let $D = \gcd(P,Q)$ and $P = DP'$, $Q = DQ'$. Then $Z \cap \bar{B}^4 = \{P' = Q' = 0\} \cap B^4$; this set is finite, and $I_u(P,Q) = I_u(P',Q')$ at each of its points.*

Proof. $Z = \{D = 0\} \cup \{P' = Q' = 0\}$. Every irreducible component of $\{D = 0\}$ lies in Z, so it does not meet S^3, and by Lemma 6.3 it does not meet $\bar{B}^4$. The coprime polynomials P', Q' have at most $\deg P' \cdot \deg Q'$ common zeros in $\mathbb{C}^2$ [Ful69, Section 5.3], and none on S^3. At a point u with $D(u) \neq 0$, D is a unit in the local ring $\mathcal{O}_u$, so $(P,Q) = (P',Q')$ in $\mathcal{O}_u$. □

Lemma 6.5 (Degree as a sum of indices)**.** *Let $\Phi\colon \bar{B}^n \to \mathbb{R}^n$ be smooth with finitely many zeros, all in the open ball. Then the degree of $\Phi/|\Phi|\colon S^{n-1} \to S^{n-1}$ is the sum of the local indices of Φ at its zeros, where the index at u is the degree of $\Phi/|\Phi|$ on a small sphere about u.*

Proof. Remove disjoint small open balls B_u about the zeros. On the compact oriented manifold $X = \bar{B}^n \setminus \bigcup B_u$ the map $\Phi/|\Phi|$ is defined, and its restriction to $\partial X = S^{n-1} \sqcup \bigsqcup(-\partial B_u)$ has degree zero, because it extends over X [Mil65, Section 5, Lemma 1]. This is the statement, input (I5). □

Proof of Theorem D. Finiteness is Lemma 6.4. The map $g = F/|F|\colon S^3 \to S^3$ satisfies $[P:Q] = h \circ g$, so by Proposition 4.3, $\mathrm{H}([P:Q]) = \deg(g)\mathrm{H}(h) = \deg g$. By Lemma 6.5, applied to $\Phi = F|_{\bar{B}^4}$, $\deg g$ is the sum of the indices of F at the points of $Z \cap B^4$. At each such point the index of the holomorphic germ F is the intersection multiplicity $I_u(P,Q)$, input (I5) [GH78, Chapter 5], [AGV85, Chapter 5].[11] This proves the formula. The sum is ≥ 0, and by Lemma 6.4 and the affine Bézout inequality [Ful69, Section 5.3],

$$\sum_{u \in Z \cap B^4} I_u(P,Q) = \sum_{u \in B^4} I_u(P',Q') \leq \sum_{u \in \mathbb{C}^2} I_u(P',Q') \leq \deg P' \cdot \deg Q' \leq \deg P \cdot \deg Q.$$

For $[z^p : w^q]$ the only common zero is 0, with $I_0(z^p, w^q) = \dim_{\mathbb{C}} \mathbb{C}[z,w]/(z^p, w^q) = pq$; for $[1:0]$, or any $[P:Q]$ with Q a nonzero constant, there are none. Since $\pi_3(S^2) \cong \mathbb{Z}$ is detected by H (Theorem C), $[P:Q]$ is null-homotopic exactly when $Z \cap B^4 = \emptyset$, every I_u being at least one. Finally $[z^m : w]$ has Hopf invariant m, and for the map $\kappa(z,w) = (\bar{z}, w)$ of degree -1 the composite $[z^m : w] \circ \kappa$ has Hopf invariant $-m$ by Proposition 4.3. □

[11] When u is a regular zero this is the observation that the real Jacobian of a holomorphic map is $|\det_{\mathbb{C}} DF|^2 > 0$, so the index is $+1$; in general one perturbs F to $F - \epsilon$ for a small regular value ϵ, whose $I_u(P,Q)$ preimages near u are all regular.

The two inequalities in the proof give an exact criterion for the upper bound.

Proposition 6.6 (Equality)**.** *Let $p = \deg P$ and $q = \deg Q$, and let $\tilde{P}, \tilde{Q}$ be the homogenisations in $\mathbb{CP}^2$. Then $\mathrm{H}([P : Q]) = pq$ if and only if P and Q have no common factor, $\tilde{P}$ and $\tilde{Q}$ have no common zero on the line at infinity, and all common zeros of P and Q in $\mathbb{C}^2$ lie in B^4.*

Proof. By Bézout's theorem in $\mathbb{CP}^2$ [Ful69, Section 5.3], two curves of degrees p and q without common component meet in exactly pq points counted with multiplicity. If $D = \gcd(P, Q)$ is not constant, the chain of inequalities above is strict at the last step. Otherwise the sum over $\mathbb{C}^2$ equals pq minus the multiplicities at infinity, and the sum over B^4 equals the sum over $\mathbb{C}^2$ exactly when no common zero lies outside $\bar{B}^4$; none lies on S^3 by hypothesis. □

Example 6.7.
(a) $[z^p : w^q]$: $\mathrm{H} = pq$. The fibre over a point $[\alpha : \beta]$ with $\alpha\beta \neq 0$ lies on the torus $|z| = r_1$, $|w| = r_2$ determined by $r_1^2 + r_2^2 = 1$ and $r_1^p|\beta| = r_2^q|\alpha|$, and there it is the set $\{p \arg z - q \arg w \equiv \arg(\alpha/\beta)\}$, the torus link of type (p, q) with $\gcd(p, q)$ components. Two such fibres have linking number pq, as Theorem 6.2 requires; Figure 3 shows the cases $(3, 1)$ and $(2, 3)$.

(b) If P and Q are homogeneous of the same degree k with no common factor, the only common zero is 0, and $I_0(P, Q) = k^2$ because the tangent cones P and Q have no common line [Ful69, Section 3.3]. So $\mathrm{H} = k^2$, in accordance with $[P : Q] = \varphi \circ h$ for the rational map $\varphi = [P : Q]$ of $\mathbb{CP}^1$ of degree k and Proposition 4.3.

(c) $[z^2 - \frac{1}{4} : w]$ has the common zeros $(\pm\frac{1}{2}, 0) \in B^4$, each of multiplicity one, so $\mathrm{H} = 2 = \deg P \cdot \deg Q$. Moving the zeros outward, $[z^2 - c^2 : w]$ has $\mathrm{H} = 2$ for $|c| < 1$ and $\mathrm{H} = 0$ for $|c| > 1$; at $|c| = 1$ the hypothesis fails, and the map is not defined everywhere.

(d) $[z^2 : zw + 1]$: the common zeros satisfy $z = 0$ and $1 = 0$, so there are none, and $\mathrm{H} = 0 < 2 \cdot 2$, although P and Q are coprime; here $\tilde{P} = z^2$ and $\tilde{Q} = zw + t^2$ meet only at $[0 : 1 : 0]$ on the line at infinity, with multiplicity four.

Corollary 6.8 (Linking as intersection)**.** *Let $[P : Q]$ be as in Theorem D, and let $[\alpha : \beta] \neq [\gamma : \delta]$ be regular values. Then the links $K_1 = \{\beta P = \alpha Q\} \cap S^3$ and $K_2 = \{\delta P = \gamma Q\} \cap S^3$ bound the holomorphic curves $C_1 = \{\beta P = \alpha Q\} \cap \bar{B}^4$ and $C_2 = \{\delta P = \gamma Q\} \cap \bar{B}^4$ in the ball, and*

$$\varepsilon \, \mathrm{lk}(K_1, K_2) = \sum_{u \in C_1 \cap C_2} I_u(\beta P - \alpha Q,\ \delta P - \gamma Q) = \mathrm{H}([P : Q]).$$

Proof. The fibres of $[P : Q]$ over the two values are K_1 and K_2, so the first and last terms agree by Theorem 6.2. Since $\alpha\delta - \beta\gamma \neq 0$, the ideal $(\beta P - \alpha Q, \delta P - \gamma Q)$ equals (P, Q), so the middle term is the sum in Theorem D. □

Remark 6.9. Corollary 6.8 is a special case of the general fact that the linking number of the boundaries of two complex curves in the ball, meeting S^3 transversally, is their intersection number in the ball, which is a sum of positive local contributions. It explains why the holomorphic Hopf invariants are non-negative: complex curves meet positively.

6.3. **Quaternionic and octonionic polynomial maps.** For $\mathbb{K} = \mathbb{H}$ or $\mathbb{O}$ there is no algebraic intersection theory, but Theorem D has an exact analogue for split maps, and it fails for general polynomial maps; this section proves Theorem G. Throughout this section $\mathbb{K} = \mathbb{H}$ or $\mathbb{O}$, $d = 4$ or 8, and $\langle u, v \rangle = \mathrm{Re}(u\bar{v})$.

A *one-sided polynomial* over $\mathbb{K}$ is a function

$$P(q) = \sum_{k=0}^{m} q^k p_k, \qquad p_k \in \mathbb{K}, \quad p_m \neq 0, \tag{6.2}$$

of degree m; the powers are well defined because a single element generates an associative subalgebra (Lemma 3.1). For a map $\Phi\colon \mathbb{R}^N \to \mathbb{R}^N$ and a compact component C of $\Phi^{-1}(0)$ that is open in $\Phi^{-1}(0)$, the *index* $\operatorname{ind}_C \Phi = \deg(\Phi, U, 0)$ is the degree of $\Phi/|\Phi|$ on the boundary of a bounded neighbourhood U of C whose closure meets no other zero; it does not depend on U. The proof of Lemma 6.5 applies verbatim with such neighbourhoods in place of small balls: if $\Phi^{-1}(0) \cap \bar{B}^N$ has finitely many components, none meeting S^{N-1}, then $\deg(\Phi/|\Phi|)$ on S^{N-1} is the sum of the indices of the components in B^N. For a regular value w near 0, the index is the number of preimages of w in U counted with the sign of the Jacobian [Mil65, Section 5].

6.3.1. *The Jacobian of a one-sided polynomial.* Write a non-real point as $q = x + yI$ with $y > 0$ and I a unit imaginary element, put $L_I = \mathbb{R}1 + \mathbb{R}I$, and write $h = h_\parallel + h_\perp$ for the orthogonal decomposition of $h \in \mathbb{K}$ along $L_I \oplus L_I^\perp$. Let $\alpha_k + i\beta_k = (x+iy)^k$ with $\alpha_k, \beta_k \in \mathbb{R}$, and

$$A(q) = \sum_{k=1}^{m} k\, q^{k-1} p_k, \qquad B(q) = \sum_{k=0}^{m} \frac{\beta_k}{y}\, p_k. \tag{6.3}$$

Lemma 6.10. *At a non-real point q, $DP_q(h) = h_\parallel A(q) + h_\perp B(q)$. At a real point x, $DP_x(h) = hA(x)$.*

Proof. For $h \in L_I$ the variation stays in the commutative subalgebra L_I, so $D(q^k)h = k\, hq^{k-1}$, and $(hq^{k-1})p_k = h(q^{k-1}p_k)$ because h, q^{k-1} and p_k lie in the subalgebra generated by I and p_k, which is associative by Artin's theorem (D4). For $h \perp L_I$ the real part x and the length y of the imaginary part of $q + th$ have zero first variation,[12] while I varies with velocity h/y. Since $q^k = \alpha_k + I\beta_k$, this gives $D(q^k)h = (\beta_k/y)h$, and multiplying on the right by p_k and summing gives the formula. At a real point every factor commutes with x, and $D(x^k)h = kx^{k-1}h$. □

Proposition 6.11. *At a non-real point with $B \neq 0$, put $s = \langle A, B\rangle$ and $t = \langle IA, B\rangle$. Then*

$$\det DP_q = |B|^{d-4}\,(s^2 + t^2) \;\geq 0. \tag{6.4}$$

If $B = 0$ the determinant vanishes, and at a real point it is $|A(x)|^d$. Consequently $\det DP \geq 0$ *everywhere, and* $\det DP > 0$ *on a dense open set.*

Proof. Right multiplication r_B by $B \neq 0$ multiplies lengths by $|B|$ (by (D1)) and preserves orientation, since its determinant does not vanish on the connected set $\mathbb{K} \setminus \{0\}$ and is 1 at $B = 1$. So if $(1, I, e_2, \dots, e_{d-1})$ is a positive orthonormal basis, $(B, IB, e_2B, \dots, e_{d-1}B)/|B|$ is a positive orthonormal basis of the target. By (D3) and Lemma 3.1, $\langle xB, yB\rangle = |B|^2\langle x, y\rangle$. By Lemma 6.10, $DP_q(e_j) = e_jB$ for $j \geq 2$, which has coordinates $|B|\delta_{jk}$ in this basis, while $DP_q(1) = A$ and $DP_q(I) = IA$. Left multiplication by I is orthogonal and satisfies $I(IX) = -X$ (Lemma 3.1), so it is skew-adjoint: $\langle IA, IB\rangle = s$ and $\langle A, IB\rangle = -t$. The matrix of DP_q is therefore block triangular, with the block $\frac{1}{|B|}\left(\begin{smallmatrix} s & t \\ -t & s \end{smallmatrix}\right)$ and the block $|B| \cdot \mathrm{Id}_{d-2}$ on the diagonal, which gives (6.4). If $B = 0$ the image of DP_q has dimension at most two. At a real point $DP_x = r_{A(x)}$ has determinant $|A(x)|^d$. Finally $\det DP$ is a real polynomial in the coordinates of q, and it is positive at large real x, where $A(x) \approx mx^{m-1}p_m \neq 0$; a non-zero polynomial does not vanish on an open set. □

[12] Since h is imaginary and orthogonal to I, $\operatorname{Re}(q + th) = x$ and $|yI + th|^2 = y^2 + t^2|h|^2$, whose derivative at $t = 0$ vanishes.

6.3.2. *The zero components.* For $z = x+iy \in \mathbb{C}$ let $U(z) = \sum_k \operatorname{Re}(z^k)p_k$ and $V(z) = \sum_k \operatorname{Im}(z^k)p_k$ in $\mathbb{K}$, so that $P(x + yI) = U + IV$ for every unit imaginary I. The *norm polynomial* of P is the real polynomial of degree $2m$

$$S_P(z) = \sum_{i,j=0}^{m} \langle p_i, p_j\rangle z^{i+j} = |U|^2 - |V|^2 + 2i\langle U, V\rangle. \tag{6.5}$$

On the real axis $S_P(x) = |P(x)|^2$.

Lemma 6.12. *The zeros of P form finitely many components, which correspond bijectively to the roots of S_P in the closed upper half-plane: a real root x corresponds to the real zero x; a root $z = x + iy$ with $y > 0$ corresponds either to the single zero $x + yJ$, $J = -UV^{-1}$, if $V(z) \neq 0$, or to the whole sphere $\{x + yI : I \in S^{d-2}\}$ of radius $|z|$ if $V(z) = 0$.*

Proof. If $P(x + yI) = 0$ with $y > 0$ then $U = -IV$, so $|U| = |V|$ and $\langle U, V\rangle = -\langle IV, V\rangle = 0$, and $S_P(z) = 0$. Conversely let $S_P(z) = 0$ with $y > 0$, so $|U| = |V|$ and $\langle U, V\rangle = 0$. If $V \neq 0$, the equation $IV = -U$ has the unique solution $I = J = -UV^{-1}$ by Lemma 3.1, with $|J| = 1$ and $\operatorname{Re} J = -\langle U, \overline{V^{-1}}\rangle = -\langle U, V\rangle/|V|^2 = 0$ by (D3); so J is a unit imaginary element and $x + yJ$ is the only zero of the form $x + yI$. If $V = 0$ then $U = 0$ and every $x + yI$ is a zero. The real case is $S_P(x) = |P(x)|^2$. Since S_P has finitely many roots, there are finitely many components, which are points or $(d - 2)$-spheres. □

Lemma 6.13. *Let $G\colon \mathbb{R}^N \to \mathbb{R}^N$ be a polynomial map with $\det DG \geq 0$, not identically zero. Then every compact component of $G^{-1}(0)$ that is open in $G^{-1}(0)$ has strictly positive index.*

Proof. Let U isolate the component C and put $\delta = \min_{\partial U} |G| > 0$. The zero set of the non-zero polynomial $\det DG$ has empty interior, so there is a point $x \in U$ near C with $\det DG_x > 0$ and $|G(x)| < \delta/2$. By the inverse function theorem G maps a small neighbourhood of x in U onto an open set $W \subset B_\delta(0)$, and by Sard's theorem W contains a regular value w. Its preimages in U are finitely many, there is at least one, and each has $\det DG > 0$. Since the segment from 0 to w avoids $G(\partial U)$, $\operatorname{ind}_C G = \deg(G, U, w) \geq 1$. □

Proposition 6.14. *Every zero component C of P has positive index, and $\sum_C \operatorname{ind}_C P = m$. If C corresponds to the root $z_0 = x_0 + iy_0$ of S_P of multiplicity ℓ, then $\operatorname{ind}_C P = \ell$ if $y_0 > 0$ and $\operatorname{ind}_C P = \ell/2$ if $y_0 = 0$; in the second case ℓ is even, and for a spherical component $\ell \geq 2$.*

Proof. Positivity follows from Lemma 6.13 and Proposition 6.11. By the multiplicativity (D1) of the norm, $|P(q)| \geq |q|^m|p_m| - \sum_{k<m} |q|^k|p_k|$, so on a large sphere the maps $q^m p_m + u\sum_{k<m} q^k p_k$, $0 \leq u \leq 1$, have no zero; with Proposition 4.8 and the orientation-preserving map r_{p_m} this gives total degree m, which is the sum of the indices.

For the multiplicities, let $w \in \mathbb{K}$ be a small regular value of P. The coefficients of S_{P-w} are close to those of S_P, so by the argument principle a small disc about z_0 contains exactly ℓ roots of S_{P-w}, counted with multiplicity. At a non-real preimage $x + yJ$ of w write $A = U_x + JV_x$, $B = V/y$, where U, V are formed for $P - w$; differentiating (6.5) with $U = -JV$ gives

$$S'_{P-w}(x + iy) = 2y\big(\langle JA, B\rangle + i\langle A, B\rangle\big), \tag{6.6}$$

which is non-zero by (6.4), since w is regular. At a real preimage x, $S_{P-w}(z) = |A(x)|^2(z - x)^2 + O((z - x)^3)$ with $A(x) \neq 0$. So each regular preimage gives a simple root of S_{P-w} in the upper half-plane, or a double real root, and by Lemma 6.12 applied to $P - w$ each such root comes from exactly one preimage (a regular value has no spherical preimage). The preimages near C are those whose roots lie in the disc: their parameters (x, y) are close to z_0, and conversely a preimage with parameters near z_0 but far from C would, in the limit, give a second zero in the conjugacy class of z_0, which does not exist when C is a point and is excluded when C is the

whole sphere. For $y_0 > 0$ take the disc in the upper half-plane: the ℓ simple roots are the ℓ preimages, all of index $+1$. For $y_0 = 0$ take a disc symmetric about the real axis: non-real roots come in conjugate pairs, each pair giving one preimage, and each real root is double, so ℓ is twice the number of preimages. At a spherical component $U = V = 0$ at z_0, and the derivative of (6.5) vanishes there, so $\ell \geq 2$. □

6.3.3. *Split maps.*

Theorem 6.15. *Let P, Q be one-sided polynomials over $\mathbb{K}$ of degrees $m, n \geq 1$, with zero components C_α of radius r_α and index μ_α, and D_β of radius s_β and index ν_β, as in Proposition 6.14. If $F(a,b) = (P(a), Q(b))$ has no zero on S^{2d-1}, then*

$$\deg \frac{F}{|F|} = \sum_{r_\alpha^2 + s_\beta^2 < 1} \mu_\alpha \nu_\beta, \qquad \mathrm{H}\Big(h_{\mathbb{K}} \circ \frac{F}{|F|}\Big) = \mathrm{H}(h_{\mathbb{K}}) \sum_{r_\alpha^2 + s_\beta^2 < 1} \mu_\alpha \nu_\beta. \tag{6.7}$$

In particular $0 \leq \deg(F/|F|) \leq mn$; the degree is mn exactly when all zeros of F lie in B^{2d} and 0 exactly when none does.

Proof. The zero components of F are the products $C_\alpha \times D_\beta$, on which $|(a,b)|^2 = r_\alpha^2 + s_\beta^2$ is constant, so each lies inside or outside the unit ball. If w, w' are small regular values of P and Q, the preimages of (w, w') near $C_\alpha \times D_\beta$ are the products of the preimages, and DF is block diagonal, so $\mathrm{ind}_{C_\alpha \times D_\beta} F = \mu_\alpha \nu_\beta$. Summing over the components in the ball gives the degree, and Proposition 4.3 gives the Hopf invariant. The last assertions follow from $\mu_\alpha, \nu_\beta \geq 1$ and $\sum \mu_\alpha \sum \nu_\beta = mn$. □

All the data in (6.7) are read off from the real polynomials S_P and S_Q: their roots in the closed upper half-plane, the multiplicities, and the moduli of the roots. For rational coefficients this is a finite exact computation.

Example 6.16. Let $P(a) = a^2 + \frac{1}{4}$ and $Q(b) = b^2 - \frac{9}{4}b + \frac{1}{2} = (b - \frac{1}{4})(b - 2)$, over $\mathbb{H}$ or $\mathbb{O}$. Then $S_P(z) = (z^2 + \frac{1}{4})^2$ has the double root $\frac{i}{2}$, and $V(\frac{i}{2}) = 0$, so the zero set of P is the sphere of imaginary elements of length $\frac{1}{2}$, of index 2.[13] The norm polynomial $S_Q = Q^2$ has the double real roots $\frac{1}{4}$ and 2, of index 1 each. The two components of the zero set of F have $r^2 + s^2 = \frac{5}{16} < 1$ and $\frac{17}{4} > 1$, so $\deg(F/|F|) = 2$ while $mn = 4$. With $P(a) = a^2 + 4$ the degree is 0.

Corollary 6.17. *For $\mathbb{K} = \mathbb{H}$ or $\mathbb{O}$ and $m, n \geq 1$, the degrees $\deg(F/|F|)$ of the split maps $F = (P, Q)$ with $\deg P = m$, $\deg Q = n$ and no zero on S^{2d-1} are exactly the integers $0, 1, \ldots, mn$, and the Hopf invariants of the maps $h_{\mathbb{K}} \circ F/|F| \colon S^{4d-1} \to S^{2d}$ obtained in this way are exactly $0, \mathrm{H}(h_{\mathbb{K}}), \ldots, mn\,\mathrm{H}(h_{\mathbb{K}})$.*[14]

Proof. By Theorem 6.15 only the values $0, \ldots, mn$ occur. Given $0 \leq k \leq mn$, write $k = c_1 + \cdots + c_m$ with $n \geq c_1 \geq \cdots \geq c_m \geq 0$. Put $s_\beta = \beta/(n+1)$ for $\beta = 1, \ldots, n$, and choose real numbers $0 < r_1 \leq \cdots \leq r_m < 1$ with $s_{c_\alpha}^2 < 1 - r_\alpha^2 < s_{c_\alpha + 1}^2$, where $s_0 = 0$ and $s_{n+1} = 1$; this is possible because $1 - r^2$ decreases from 1 to 0 as r increases from 0 to 1, and the intervals are taken in decreasing order as α increases. Let $P(a) = \prod_\alpha (a - r_\alpha)$ and $Q(b) = \prod_\beta (b - s_\beta)$, with real coefficients. Their norm polynomials are $P(z)^2$ and $Q(z)^2$, whose roots in the closed

[13] Every unit imaginary I satisfies $I^2 = -1$, so $(\frac{1}{2}I)^2 = -\frac{1}{4}$; over $\mathbb{C}$ the same polynomial has the two simple zeros $\pm\frac{i}{2}$, which the sphere joins together. A small perturbation $a^2 + \frac{1}{4} - w$ with w non-real has exactly two zeros, the square roots of $w - \frac{1}{4}$ in the slice of w, which is how the index 2 of Proposition 6.14 appears.

[14] For $\mathbb{K} = \mathbb{C}$ the same set of values is attained by holomorphic maps, by Theorem D; the corollary shows that splitting the variables over $\mathbb{H}$ or $\mathbb{O}$ loses nothing in the range, although it loses the negative values that the general maps of Proposition 6.19 attain.

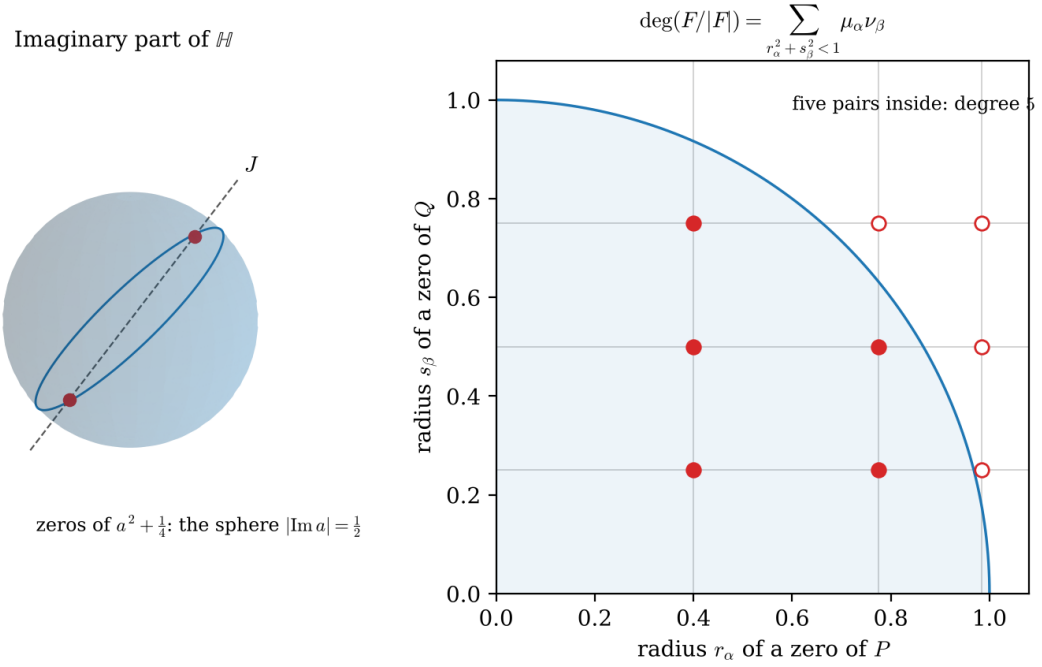


FIGURE 4. Left: the zero set of $a^2 + \frac{1}{4}$ over $\mathbb{H}$, drawn in $\mathrm{Im}\,\mathbb{H} = \mathbb{R}^3$, is the sphere of imaginary quaternions of length $\frac{1}{2}$ (Example 6.16); the imaginary parts of the two preimages of a small regular value w are marked, on the line through the unit J of the slice of w. Right: the construction in the proof of Corollary 6.17 for $m = n = 3$ and $(c_1, c_2, c_3) = (3, 2, 0)$; each point is a zero component $\{r_\alpha\} \times \{s_\beta\}$ of $F = (P, Q)$, and the five inside the unit quarter-circle give $\deg(F/|F|) = 5$.

upper half-plane are the real numbers r_α and s_β, of multiplicity twice the multiplicity in P or Q; by Proposition 6.14 the zero components are the points r_α and s_β, of index equal to their multiplicity. The pairs with $r_\alpha^2 + s_\beta^2 < 1$ are those with $\beta \le c_\alpha$, and no pair has $r_\alpha^2 + s_\beta^2 = 1$, so (6.7) gives $\deg(F/|F|) = \sum_\alpha c_\alpha = k$. The Hopf invariants follow from Proposition 4.3. □

Theorem 6.18. *Let P, Q be as in Theorem 6.15 and let $R\colon \mathbb{K} \to \mathbb{K}$ be any polynomial map in the real coordinates. The map $T(a, b) = (P(a), Q(b) + R(a))$ is proper of degree mn, its zero set has finitely many components, each of positive index, and if none of them meets S^{2d-1} then $\deg(T/|T|)$ is the sum of the indices of the components in B^{2d}, so that $0 \le \deg(T/|T|) \le mn$, with equality on the left exactly when T has no zero in the ball.*

Proof. The differential is block triangular, $\det DT = \det DP \cdot \det DQ \ge 0$, and it is not identically zero by Proposition 6.11. If $T(a, b)$ is bounded then a is bounded since P is proper, so $R(a)$ is bounded and b is bounded since Q is proper. The zeros of $T_u = (P, Q + uR)$, $0 \le u \le 1$, have a in the compact set $P^{-1}(0)$ and b bounded uniformly in u, so on a large sphere T is homotopic to $P \times Q$, of degree mn by Theorem 6.15. The zero set is real algebraic and compact, so it has finitely many connected components by (I8) [BCR98, Theorem 2.4.4], and Lemma 6.13 gives positive indices. □

6.3.4. *General polynomial maps.* Polynomials with constants between the variable factors are much more general, since they include conjugation. Let $e_1, \dots, e_{d-1}$ be an orthonormal basis of

$\operatorname{Im}\mathbb{K}$ as in Appendix A. Then

$$\bar{a} = -\frac{1}{d-2}\Big(a + \sum_{j=1}^{d-1}(e_j a)e_j\Big), \qquad \operatorname{Re} a = \frac{(d-3)a - \sum_j (e_j a)e_j}{2(d-2)}. \tag{6.8}$$

Both sides are real linear in a, so it suffices to check them at $a = 1$, where $\sum_j (e_j 1)e_j = -(d-1)$, and at $a = e_k$, where the term $j = k$ is $-e_k$ and the other $d-2$ terms are $(e_j e_k)e_j = e_k$, computed in the associative subalgebra generated by e_j and e_k.

Proposition 6.19. *For every integer k there is a polynomial map $G\colon \mathbb{K}^2 \to \mathbb{K}^2$, with constants between the variable factors, whose only zero is the origin and with $\deg(G/|G|) = k$. For $k = -1$ one can take $G(a,b) = (\bar{a}, b)$ or the coupled map $(\bar{a} + ba, b)$.*

Proof. By (6.8), $\bar{a}$ and the real coordinates $x_0 = \operatorname{Re} a$, $x_j = \operatorname{Re}(-e_j a)$ of a are polynomials in this sense. The map $(\bar{a}, b)$ is linear with determinant $(-1)^{d-1} = -1$;[15] for $(\bar{a} + uba, b)$ the second coordinate forces $b = 0$ and then $a = 0$, so the degree stays -1 for $0 \le u \le 1$. For $k \ge 1$ put $z = x_0 + x_1 e_1$ and $G_k^{\pm}(a,b) = (z^k + \sum_{j\ge 2} x_j e_j,\ b)$, respectively with $\bar{z}^k$: in real coordinates these are the complex maps z^k and $\bar{z}^k$ times the identity, of degrees k and $-k$. Finally $G_0(a,b) = (x_0^2 + \sum_{j\ge 1} x_j e_j,\ b)$ has only the zero 0, and values with negative real part near 0 are not attained, so its degree is 0. □

So there is no formula of the type of Theorem D with positive contributions for general polynomial maps over $\mathbb{H}$ or $\mathbb{O}$; the positive results above use that each coordinate is a one-sided polynomial in one variable, with at most a triangular coupling. The one-variable results have a long history. That the zeros of a one-sided quaternionic polynomial are isolated points and spheres of imaginary elements is due to Niven [Niv41] and to Gordon and Motzkin [GM65], and the octonionic case to Serôdio [Ser07]; that a polynomial with a single leading term has a zero is the fundamental theorem of algebra of Eilenberg and Niven [EN44], proved there by the degree argument used in Proposition 6.14. One-sided polynomials are the slice regular polynomials of Gentili, Stoppato and Struppa [GSS13], and the spherical zeros are the phenomenon studied by Ghiloni and Perotti [GP11]; Sakkalis [Sak20] relates the degrees of polynomial maps of normed algebras to their leading terms. In that literature S_P appears as the companion, or symmetrised, polynomial $P * P^c$. The proofs above do not depend on these sources; what the Hopf invariant needs from them is the topological index of each component, the sign of the Jacobian (6.4), and the formula (6.7) with its triangular extension, and we have not found these stated in the references cited.

Question 6.20. For which classes of polynomial maps $F = (P, Q)\colon \mathbb{K}^2 \to \mathbb{K}^2$ in two variables, beyond the triangular maps of Theorem 6.18 (for instance for a fixed notion of slice regularity in several variables), is $\det DF \ge 0$, so that every zero component has positive index and $\deg(F/|F|)$ is a positive count of the zeros in B^{2d}?

7. The metric geometry of the Hopf fibrations

This section proves Theorem F. In Sections 7.1 and 7.2, $\mathbb{K} = \mathbb{C}$ or $\mathbb{H}$, $d = \dim_{\mathbb{R}} \mathbb{K} \in \{2, 4\}$; Section 7.3 treats all four algebras, including $\mathbb{O}$. Throughout, S^{2d-1} carries the round metric of radius one. For $u = (a,b)$ and $v = (a',b')$ in $\mathbb{K}^2$ we write $\langle u, v\rangle = \bar{a}a' + \bar{b}b' \in \mathbb{K}$, so that $\operatorname{Re}\langle u, v\rangle$ is the Euclidean inner product and $\langle u, u\rangle = |u|^2$.

[15] Conjugation fixes the real axis and reverses the $d-1$ imaginary directions; the second factor is the identity.

7.1. **The Hopf map as a map of matrices.** Regard $u \in \mathbb{K}^2$ as a column vector and let $u^* = (\bar a, \bar b)$. The matrix uu^* is Hermitian, and

$$2uu^* - |u|^2 I = \begin{pmatrix} |a|^2 - |b|^2 & 2a\bar b \\ 2b\bar a & |b|^2 - |a|^2 \end{pmatrix}. \tag{7.1}$$

The traceless Hermitian 2×2 matrices over $\mathbb{K}$ form the real vector space $\mathcal{H}_0 = \{X_{t,x} = \left(\begin{smallmatrix} t & x \\ \bar x & -t \end{smallmatrix}\right) : t \in \mathbb{R},\ x \in \mathbb{K}\}$ of dimension $d+1$, and $X_{t,x} \mapsto (x,t)$ is an isometry onto $\mathbb{K}\times\mathbb{R}$ for the norm $\|X\|^2 = \frac12 \operatorname{Re}\operatorname{tr}(X^2) = t^2 + |x|^2$. By (7.1), $h_{\mathbb{K}}(u)$ corresponds to $2uu^* - I$ for $|u| = 1$.

Lemma 7.1. *Let $G = \mathrm{U}(2)$ for $\mathbb{K} = \mathbb{C}$ and $G = \mathrm{Sp}(2)$ for $\mathbb{K} = \mathbb{H}$, acting on $\mathbb{K}^2$ by matrix multiplication. Then G acts transitively on S^{2d-1} by isometries, $\mathrm{Ad}(g)X = gXg^*$ is a linear isometry of $\mathcal{H}_0$, and $h_{\mathbb{K}}(gu) = \mathrm{Ad}(g)\, h_{\mathbb{K}}(u)$.*

Proof. Transitivity on unit vectors is the statement that every unit vector is the first column of a unitary, respectively symplectic, matrix, which follows from Gram–Schmidt over $\mathbb{K}$. The identity $(gu)(gu)^* = g(uu^*)g^*$ and $gg^* = I$ give the equivariance. For the isometry, $\operatorname{Re}\operatorname{tr}(XY) = \operatorname{Re}\operatorname{tr}(YX)$ for quaternionic matrices, because $\operatorname{Re}(xy) = \operatorname{Re}(yx)$ for quaternions, so $\operatorname{Re}\operatorname{tr}\big((gXg^*)^2\big) = \operatorname{Re}\operatorname{tr}(gX^2g^*) = \operatorname{Re}\operatorname{tr}(X^2)$. □

Proposition 7.2. *For unit vectors $u, v \in \mathbb{K}^2$,*

$$h_{\mathbb{K}}(u) \cdot h_{\mathbb{K}}(v) = 2\,|\langle u, v\rangle|^2 - 1.$$

Proof. With $A = uu^*$ and $B = vv^*$, the inner product of $2A - I$ and $2B - I$ in $\mathcal{H}_0$ is $\frac12 \operatorname{Re}\operatorname{tr}\big((2A - I)(2B - I)\big) = 2\operatorname{Re}\operatorname{tr}(AB) - \operatorname{tr} A - \operatorname{tr} B + 1 = 2\operatorname{Re}\operatorname{tr}(AB) - 1$. Moving the column u around the trace, $\operatorname{Re}\operatorname{tr}(u\,u^*vv^*) = \operatorname{Re}\big((u^*v)(v^*u)\big) = |\langle u, v\rangle|^2$, since $v^*u = \overline{u^*v}$. □

7.2. **Riemannian submersion.**

Proof of Theorem F, first part. By Lemma 7.1 and the chain rule, $Dh_{\mathbb{K}}(gu) \circ Dg = \mathrm{Ad}(g) \circ Dh_{\mathbb{K}}(u)$; since g and $\mathrm{Ad}(g)$ are isometries and g maps fibres to fibres, it suffices to check at $u = (1,0)$ that $Dh_{\mathbb{K}}$ vanishes on the tangent space of the fibre and multiplies the length of vectors orthogonal to it by 2. The tangent space to S^{2d-1} at $(1,0)$ is $\{(\alpha,\beta) : \operatorname{Re}\alpha = 0\}$. The fibre through $(1,0)$ is $\{(q,0) : |q| = 1\}$, with tangent space $\operatorname{Im}\mathbb{K} \times 0$, and its orthogonal complement is $0 \times \mathbb{K}$. From $h_{\mathbb{K}}(1+\alpha,\beta) = \big(2(1+\alpha)\bar\beta,\ |1+\alpha|^2 - |\beta|^2\big)$,

$$Dh_{\mathbb{K}}(1,0)(\alpha,\beta) = \big(2\bar\beta,\ 2\operatorname{Re}\alpha\big) = (2\bar\beta, 0),$$

which is zero on the fibre directions and has length $2|\beta|$ on horizontal vectors. Hence $\frac12 h_{\mathbb{K}} \colon S^{2d-1} \to S^d(\frac12)$ is a Riemannian submersion. □

Proposition 7.3 (Parallel fibres)**.** *Let $u, v \in S^{2d-1}$ and let $\psi \in [0,\pi]$ be the angle between $h_{\mathbb{K}}(u)$ and $h_{\mathbb{K}}(v)$. Then every point of the fibre through u has distance exactly $\psi/2$ from the fibre through v, and this is the distance between $\frac12 h_{\mathbb{K}}(u)$ and $\frac12 h_{\mathbb{K}}(v)$ in $S^d(\frac12)$.*

Proof. The fibre through v is $\{vq : |q| = 1\}$ (Proposition 3.7). The spherical distance from u to vq is $\arccos\operatorname{Re}\langle u, vq\rangle = \arccos\operatorname{Re}(\langle u, v\rangle q)$, and the maximum of $\operatorname{Re}(\langle u, v\rangle q)$ over unit q is $|\langle u, v\rangle|$. So the distance from u to the fibre is $\theta = \arccos|\langle u, v\rangle| \in [0, \pi/2]$, which depends only on the fibres. By Proposition 7.2, $\cos\psi = 2\cos^2\theta - 1 = \cos 2\theta$, so $\theta = \psi/2$, which is the distance in the sphere of radius $\frac12$. □

7.3. **The octonionic Hopf map.** The proof above uses the transitive groups $\mathrm{U}(2)$ and $\mathrm{Sp}(2)$, which have no octonionic analogue: there is no group of $\mathbb{O}$-linear maps of $\mathbb{O}^2$, since $\mathbb{O}$ is not associative. The following argument uses only the lines L_x of Proposition 3.3 and Lemma 3.1, and it applies to all four algebras.

Lemma 7.4. *Let $x \in \mathbb{K}$. The orthogonal complement of the line $L_x = \{(xc, c) : c \in \mathbb{K}\}$ in $\mathbb{K}^2$ is $L_x^\perp = \{(u, -\bar{x}u) : u \in \mathbb{K}\}$, and the orthogonal projection of $(y, 0)$ onto $L_x^\perp$ is $(u, -\bar{x}u)$ with $u = y/(1 + |x|^2)$; its length is $|y|/\sqrt{1 + |x|^2}$.*

Proof. By (D3), $\langle (u, w), (xc, c)\rangle = \langle u, xc\rangle + \langle w, c\rangle = \langle \bar{x}u + w, c\rangle$, which vanishes for all c exactly when $w = -\bar{x}u$. Both L_x and the space $\{(u, -\bar{x}u)\}$ have real dimension d, so the second is $L_x^\perp$. For $u = y/(1 + |x|^2)$ and $c = \bar{x}u$,

$$(xc, c) + (u, -\bar{x}u) = \big(x(\bar{x}u) + u,\ 0\big) = \big((|x|^2 + 1)u, 0\big) = (y, 0)$$

by Lemma 3.1, which is the orthogonal decomposition of $(y, 0)$. Finally $|(u, -\bar{x}u)|^2 = |u|^2(1 + |x|^2) = |y|^2/(1 + |x|^2)$ by (D1). □

Proposition 7.5. *For each of $\mathbb{K} = \mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$ the map $\frac{1}{2}h_{\mathbb{K}} \colon S^{2d-1} \to S^d(\frac{1}{2})$ is a Riemannian submersion whose fibres are great $(d-1)$-spheres. In particular the octonionic Hopf map $S^{15} \to S^8(\frac{1}{2})$ is a Riemannian submersion with totally geodesic fibres S^7.*

Proof. Let $p \in S^{2d-1}$ with second coordinate nonzero, so that $p \in L_x$ for $x \in \mathbb{K}$ (Proposition 3.3). The fibre through p is $L_x \cap S^{2d-1}$, so the vertical space is $V_p = L_x \cap p^\perp$, and since $L_x = V_p \oplus \mathbb{R}p$ the horizontal space is $H_p = V_p^\perp \cap p^\perp = L_x^\perp$. The derivative $Dh_{\mathbb{K}}(p)$ vanishes on V_p, because $h_{\mathbb{K}}$ is constant on the fibre.

Consider $F \colon \mathbb{K} \times S^{d-1} \to S^{2d-1}$, $F(x, c) = (xc, c)/\rho$ with $\rho = \sqrt{1 + |x|^2}$, so that $p = F(x, c)$ for some unit c and, by the computation in the proof of Theorem A, $h_{\mathbb{K}} \circ F(x, c) = s(x)$. For $\xi \in \mathbb{K}$,

$$DF(x, c)(\xi, 0) = \frac{(\xi c, 0)}{\rho} - \frac{\langle x, \xi\rangle}{\rho^3}(xc, c) \in \frac{(\xi c, 0)}{\rho} + L_x.$$

This vector is tangent to S^{2d-1} at p, and so are its components in L_x and in $L_x^\perp \subset p^\perp$; the first lies in V_p. By Lemma 7.4 its horizontal component $w(\xi)$ has length $|\xi c|/(\rho \cdot \rho) = |\xi|/(1 + |x|^2)$, and $Dh_{\mathbb{K}}(p)\, w(\xi) = D(h_{\mathbb{K}} \circ F)(x, c)(\xi, 0) = Ds(x)\xi$. The map $\xi \mapsto w(\xi)$ is linear and injective, hence onto the d-dimensional space H_p. Differentiating (3.3),

$$Ds(x)\xi = \frac{2}{1 + |x|^2}\Big(\xi - \frac{2\langle x, \xi\rangle}{1 + |x|^2}\, x,\ \frac{2\langle x, \xi\rangle}{1 + |x|^2}\Big),$$

$$|Ds(x)\xi|^2 = \frac{4}{(1 + |x|^2)^2}\Big(|\xi|^2 - \frac{4\langle x, \xi\rangle^2}{1 + |x|^2} + \frac{4\langle x, \xi\rangle^2(|x|^2 + 1)}{(1 + |x|^2)^2}\Big),$$

that is, $|Ds(x)\xi| = 2|\xi|/(1 + |x|^2) = 2|w(\xi)|$. So $Dh_{\mathbb{K}}(p)$ multiplies the length of every horizontal vector by 2, and $\frac{1}{2}h_{\mathbb{K}}$ restricted to H_p is a linear isometry onto the tangent space of $S^d(\frac{1}{2})$.

For $p = (a, 0)$ use the isometry $\sigma(a, b) = (b, a)$ of S^{2d-1}: by (D2), $h_{\mathbb{K}} \circ \sigma = r \circ h_{\mathbb{K}}$ with the isometry $r(y, t) = (\bar{y}, -t)$ of S^d, and $\sigma(p)$ has nonzero second coordinate. The fibres are great spheres by Theorem A, and great spheres are totally geodesic. □

Proposition 7.6. *For $\mathbb{K} = \mathbb{O}$, and in fact for each $\mathbb{K}$, every point of the fibre over $p \in S^d$ has distance exactly $\psi/2$ from the fibre over q, where ψ is the angle between p and q.*

Proof. Write $\pi = \frac{1}{2}h_{\mathbb{K}}$ and d_B for the distance on $S^d(\frac{1}{2})$, so $d_B(\pi(u), \pi(v)) = \psi/2$. Since $|D\pi(w)| \le |w|$ for every tangent vector w (the vertical part of w is killed and the horizontal part is preserved), π does not increase the lengths of curves, and every curve from u to the fibre over

q has length at least $\psi/2$. Conversely let γ be a minimising geodesic of $S^d(\frac{1}{2})$ from $\pi(u)$ to $\pi(v)$. Because the fibres are compact, γ has a horizontal lift starting at u: the horizontal lift of the velocity field is a smooth vector field along the fibres over γ, whose integral curves exist for all times. The lift has the length $\psi/2$ of γ and ends on the fibre over q. □

Remark 7.7 (Volumes). For a Riemannian submersion with fibres of constant volume, the volume of the total space is the product of the volumes of base and fibre. The identities

$$2\pi^2 = 2\pi \cdot \pi, \qquad \frac{\pi^4}{3} = 2\pi^2 \cdot \frac{\pi^2}{6}$$

are $\operatorname{vol} S^3 = \operatorname{vol} S^1 \cdot \operatorname{vol} S^2(\frac{1}{2})$ and $\operatorname{vol} S^7 = \operatorname{vol} S^3 \cdot \operatorname{vol} S^4(\frac{1}{2})$. By Proposition 7.5 the octonionic identity $\frac{\pi^8}{2520} = \frac{\pi^4}{3} \cdot \frac{\pi^4}{840}$, that is $\operatorname{vol} S^{15} = \operatorname{vol} S^7 \cdot \operatorname{vol} S^8(\frac{1}{2})$, has the same meaning. The full symmetry group of the octonionic Hopf fibration is Spin(9) [GWZ86], which acts transitively on S^{15} but not through $\mathbb{O}$-linear maps; Proposition 7.5 does not need it.

7.4. **Clifford tori.** For $a, b > 0$ with $a^2 + b^2 = 1$ let $T_{a,b} = S^1(a) \times S^1(b) = \{(z, w) : |z| = a,\ |w| = b\} \subset S^3$. It is the preimage under h of the circle of latitude $\{x_3 = a^2 - b^2\}$, and every point of S^3 with $zw \neq 0$ lies on exactly one of these tori.

Proposition 7.8. *The torus $T_{a,b}$ is flat, with the product metric of two circles of radii a and b. In the coordinates $(z, w) = (ae^{i\alpha}, be^{i\beta})$ the fibres of h on $T_{a,b}$ are the lines $\beta - \alpha = \text{const}$, which in the arc-length coordinates $(a\alpha, b\beta)$ are the closed geodesics of slope b/a and length 2π.*

Proof. The embedding $(\alpha, \beta) \mapsto (ae^{i\alpha}, be^{i\beta})$ has $|\partial_\alpha| = a$, $|\partial_\beta| = b$ and $\partial_\alpha \perp \partial_\beta$, which is the product metric. The fibre through $(ae^{i\alpha_0}, be^{i\beta_0})$ is $\{(ae^{i(\alpha_0+t)}, be^{i(\beta_0+t)})\}$. Straight lines in a flat torus are geodesics, and the length is $2\pi\sqrt{a^2 + b^2} = 2\pi$. □

At $a = b = 1/\sqrt{2}$ the fibres have slope one; this is the Clifford torus, the preimage of the equator. As $a \to 0$ the tori shrink to the fibre $\{z = 0\}$ over the south pole, and as $b \to 0$ to the fibre $\{w = 0\}$ over the north pole.

7.5. **The unit tangent bundle of the two-sphere.** Identify $\mathbb{R}^3$ with $\operatorname{Im}\mathbb{H}$ and S^3 with the unit quaternions. The map $\rho(q)(v) = qv\bar{q}$ is a surjective homomorphism $S^3 \to \mathrm{SO}(3)$ with kernel $\{\pm 1\}$ [Bae02, Section 2.3], so $\mathrm{SO}(3) \cong S^3/\{\pm 1\} = \mathbb{RP}^3$.

Proposition 7.9. *Let $T^1S^2 = \{(p, v) \in S^2 \times S^2 : p \perp v\}$ be the unit tangent bundle. The map $\Psi\colon S^3 \to T^1S^2$, $\Psi(q) = (qi\bar{q},\ qj\bar{q})$, is a two-fold covering that identifies T^1S^2 with $\mathbb{RP}^3$. Its composite with the projection to S^2 is $h'(q) = qi\bar{q}$, which is the Hopf map up to isometries of S^3 and of S^2, and the circle bundle $T^1S^2 \to S^2$ is the quotient of the Hopf bundle by the antipodal action $q \mapsto -q$ on its fibres.*

Proof. A point (p, v) of T^1S^2 determines the positive orthonormal frame $(p, v, p \times v)$, and $\mathrm{SO}(3)$ acts simply transitively on such frames, so $(p, v) \mapsto$ the rotation taking (i, j, k) to the frame is a diffeomorphism $T^1S^2 \cong \mathrm{SO}(3)$, and Ψ corresponds to ρ. For $q = z + wj$ with $z, w \in \mathbb{C}$ a direct computation, using $ji = -ij$ and $wj = j\bar{w}$, gives

$$qi\bar{q} = (|z|^2 - |w|^2)\, i + 2\operatorname{Im}(zw)\, j - 2\operatorname{Re}(zw)\, k,$$

which is $h(z, \bar{w}) = (2zw, |z|^2 - |w|^2)$ followed by a linear isometry of $\mathbb{R}^3$. The fibres of h' are the circles $\{qe^{i\theta}\}$, and $\Psi(qe^{i\theta}) = (qi\bar{q},\ qe^{2i\theta}j\bar{q})$, so as θ runs over $[0, \pi]$ the tangent vector turns once around $p = qi\bar{q}$; the points q and $-q$ of a fibre of h' give the same point of T^1S^2. □

Remark 7.10. The Euler number of the Hopf bundle is ± 1 and that of $T^1S^2 \to S^2$ is $\chi(S^2) = 2$, in accordance with the quotient by ± 1, which doubles the Euler class of an oriented circle bundle. The total space of T^1S^2 is $\mathbb{RP}^3$, not S^3; its fundamental group is $\mathbb{Z}/2$, while S^3 is simply connected. Theorem F follows from Propositions 7.3, 7.5, 7.6, 7.8 and 7.9.

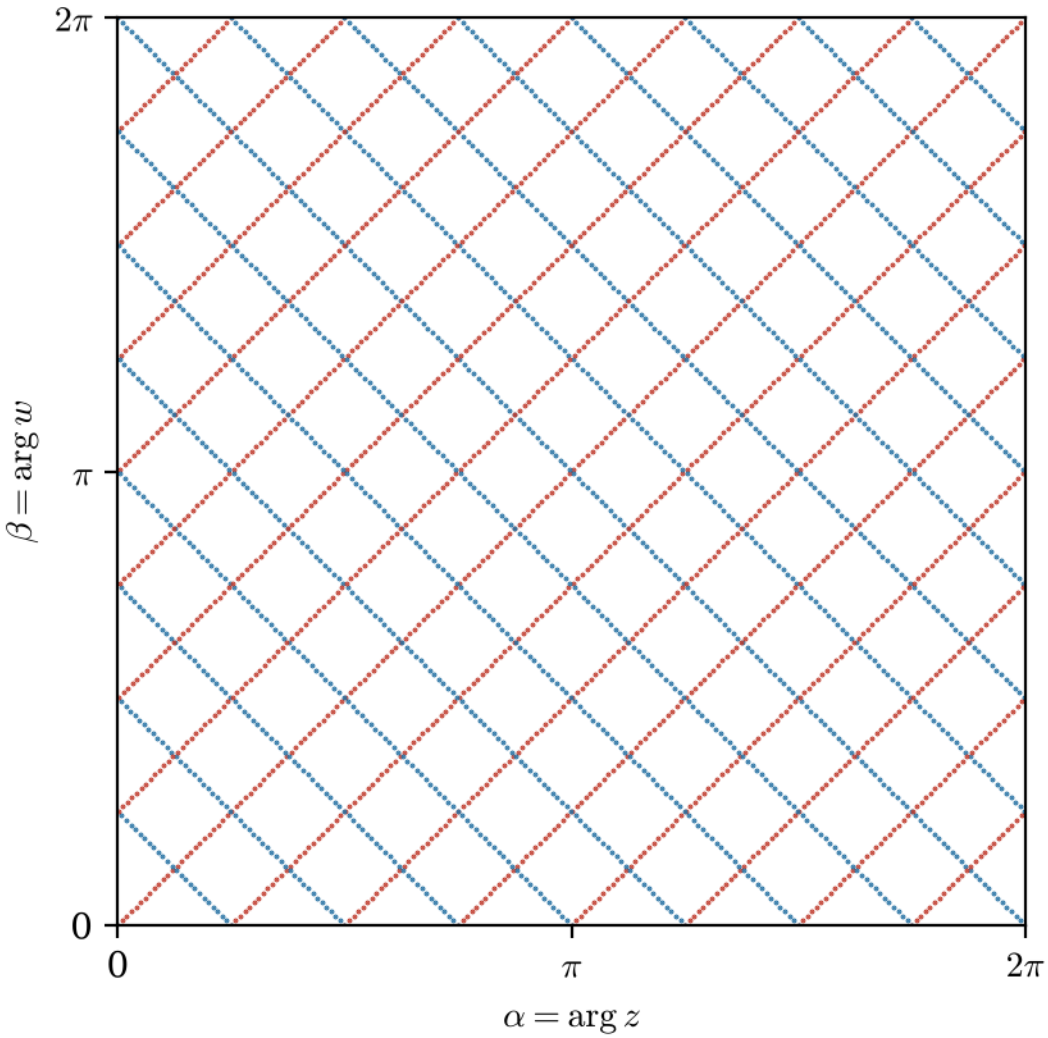


FIGURE 5. The flat torus $T_{a,b}$ unrolled in arc-length coordinates, for $a = 0.6$, $b = 0.8$. The Hopf fibres are the lines of slope b/a (solid), and the fibres of $(z, w) \mapsto h(\bar{z}, w)$ are the lines of slope $-b/a$ (dashed). After stereographic projection the two families become the two families of Villarceau circles (Section 8).

8. Villarceau circles and Villarceau spheres

This section proves Theorem E and Villarceau's theorem. The argument is the same for $d = 2, 4, 8$: one fibre is computed explicitly, and the others are obtained from it by isometries that commute with stereographic projection. For octonions the only point that needs comment is that these isometries are left multiplications, which are orthogonal by (D1).

8.1. **Stereographic projection.** Fix $d \in \{2, 4, 8\}$ and a unit vector $\varepsilon \in \mathbb{K}$ with $\varepsilon \perp 1$. For $v \in \mathbb{K}$ write $v_\varepsilon = \langle v, \varepsilon \rangle = \mathrm{Re}(v\bar{\varepsilon})$ and $\pi(v) = v - v_\varepsilon \varepsilon \in \varepsilon^\perp$, the orthogonal projection onto the hyperplane $\varepsilon^\perp \subset \mathbb{K}$. Stereographic projection from the point $N = (0, \varepsilon)$ of S^{2d-1} onto its tangent

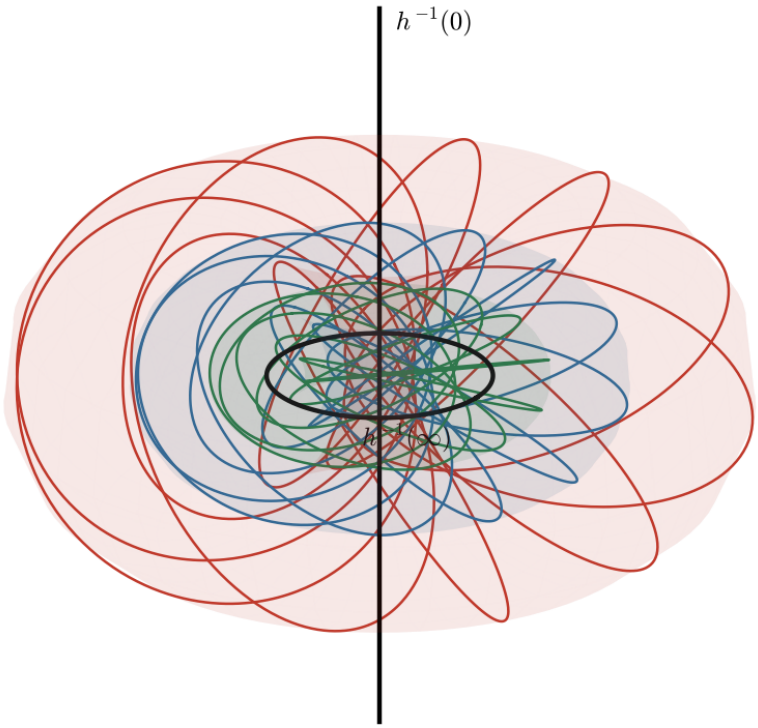


FIGURE 6. Nested tori $T_{a,b}$ in $\mathbb{R}^3$ after stereographic projection, each with some of its Hopf fibres; the vertical axis together with ∞ is the fibre over one pole, and the unit circle in the horizontal plane is the fibre over the other.

hyperplane at the antipode, translated to the origin, is

$$\sigma\colon S^{2d-1}\setminus\{N\}\to\mathbb{K}\times\varepsilon^{\perp}=\mathbb{R}^d\times\mathbb{R}^{d-1},\qquad \sigma(A,B)=\frac{\big(A,\ \pi(B)\big)}{1-B_\varepsilon}. \tag{8.1}$$

It is a conformal diffeomorphism, it maps round spheres not through N to round spheres [Lyo03], and it commutes with every isometry of $\mathbb{K}^2$ that fixes N and preserves the decomposition $\mathbb{K}\times\mathbb{K}$. For $d=2$ and $\varepsilon=i$,

$$\sigma(z,w)=\frac{\big(z,\ \operatorname{Re}w\big)}{1-\operatorname{Im}w}\in\mathbb{C}\times\mathbb{R}=\mathbb{R}^3.$$

For $0<a<1$ put $b=\sqrt{1-a^2}$ and $\Sigma_a=S^{d-1}(a)\times S^{d-1}(b)=\{(A,B):|A|=a,\ |B|=b\}\subset S^{2d-1}$. It does not contain N. Every point (A,B) of S^{2d-1} with $A\neq 0\neq B$ lies on exactly one Σ_a, namely $a=|A|$, and $\Sigma_a=h_{\mathbb{K}}^{-1}\big(\{x_{d+1}=a^2-b^2\}\big)$ is the preimage of a sphere of latitude.

Lemma 8.1. *Let $R = 1/a$ and $r = b/a$, so that $R^2 - r^2 = 1$. Then σ maps Σ_a diffeomorphically onto the tube*

$$T_{R,r} = \big\{(x,y) \in \mathbb{R}^d \times \mathbb{R}^{d-1} : (|x| - R)^2 + |y|^2 = r^2\big\}.$$

Proof. Let $(A,B) \in \Sigma_a$ and $\beta = B_\varepsilon \in [-b,b]$, so $|\pi(B)|^2 = b^2 - \beta^2$. For $(x,y) = \sigma(A,B)$ we have $|x| = a/(1-\beta)$ and $|y|^2 = (b^2-\beta^2)/(1-\beta)^2$, hence

$$a^2(1-\beta)^2\big((|x|-R)^2+|y|^2\big) = \big(a^2-(1-\beta)\big)^2 + a^2(b^2-\beta^2) = (\beta-b^2)^2 + a^2b^2 - a^2\beta^2 = b^2(1-\beta)^2,$$

using $a^2 - 1 = -b^2$ and $a^2 + b^2 = 1$. So $(|x|-R)^2 + |y|^2 = b^2/a^2 = r^2$. The restriction of σ to Σ_a is an injective immersion between compact connected manifolds of the same dimension $2d-2$, so it is a diffeomorphism onto $T_{R,r}$ by invariance of domain. □

For $d = 2$, $T_{R,r}$ is the torus obtained by rotating the circle $(x_1 - R)^2 + x_3^2 = r^2$ of the (x_1, x_3)-plane about the x_3-axis, a ring torus because $R > r$. For $d = 4, 8$ it is the boundary of the tubular neighbourhood of radius r of the round sphere $S^{d-1}(R) \subset \mathbb{R}^d \times 0$, diffeomorphic to $S^{d-1} \times S^{d-1}$.

8.2. **The fibres on a product of spheres.** By Corollary 3.5 the fibres of $h_{\mathbb{K}}$ contained in Σ_a are those over the points $s(x)$ with $|x| = a/b$. Writing $x = (a/b)u$ with $|u| = 1$ and $c = bc'$, the fibre over $s(x)$ is

$$F_u = \big\{\big(a\,(uc'),\ b\,c'\big) : |c'| = 1\big\}, \qquad u \in S^{d-1}, \tag{8.2}$$

where we used $((a/b)u)(bc') = a(uc')$, valid because a, b are real. In particular $F_1 = \{(ac, bc) : |c| = 1\}$.

Proposition 8.2 (The base fibre)**.** *Let $\Pi_1 = \{(x,\, r\,\pi(x)) : x \in \mathbb{K}\} \subset \mathbb{R}^d \times \mathbb{R}^{d-1}$, a d-dimensional linear subspace. Then $\sigma(F_1)$ is the round $(d-1)$-sphere in Π_1 with centre $(r\varepsilon, 0)$ and radius R.*

Proof. For $c \in S^{d-1}$ put $t = c_\varepsilon$. Then $\sigma(ac, bc) = (x,y)$ with $x = ac/(1-bt)$ and $y = b\pi(c)/(1-bt) = (b/a)\pi(x)$, so $\sigma(F_1) \subset \Pi_1$. For $\rho \in \mathbb{R}$,

$$(1-bt)^2\big(|x - \rho\varepsilon|^2 + |y|^2\big) = a^2 - 2a\rho t(1-bt) + \rho^2(1-bt)^2 + b^2(1-t^2),$$

and for $\rho = r = b/a$ the right-hand side equals $(1+r^2)(1-bt)^2$, as one checks by comparing the coefficients of 1, t and t^2:

$$1 + r^2, \qquad -2a\rho - 2b\rho^2 = -2b(1+r^2), \qquad -b^2 + 2ab\rho + b^2\rho^2 = b^2(1+r^2).$$

Hence $|x - r\varepsilon|^2 + |y|^2 = 1 + r^2 = R^2$. The centre $(r\varepsilon, 0)$ lies in Π_1, since $\pi(\varepsilon) = 0$. So $\sigma(F_1)$ lies on the round $(d-1)$-sphere S_1^+ of Π_1 with this centre and radius, and it fills it, being a compact $(d-1)$-manifold embedded in the connected $(d-1)$-manifold S_1^+. □

Proof of Theorem E. For $u \in S^{d-1}$ the map $\Lambda_u(A,B) = (uA, B)$ is an isometry of $\mathbb{K}^2$, since ℓ_u is orthogonal, and it fixes N and preserves Σ_a. By (8.1), $\sigma \circ \Lambda_u = \lambda_u \circ \sigma$ with $\lambda_u(x,y) = (ux, y)$, an isometry of $\mathbb{R}^d \times \mathbb{R}^{d-1}$. By (8.2), $\Lambda_u(F_1) = \{(u(ac), bc)\} = F_u$, because $u(ac) = a(uc)$. Hence

$$\sigma(F_u) = \lambda_u\big(\sigma(F_1)\big)$$

is the round $(d-1)$-sphere of radius R with centre $(ur\varepsilon, 0)$, at distance r from the origin, in the d-dimensional linear subspace $\Pi_u = \lambda_u(\Pi_1)$. The fibres F_u partition Σ_a by Proposition 3.3, and $\sigma|_{\Sigma_a}$ is a bijection onto $T_{R,r}$ (Lemma 8.1), so the spheres $\sigma(F_u)$ partition $T_{R,r}$. For $d = 2$ the statements about the two families of Villarceau circles are proved in Section 8.3 below. □

We call the spheres $\sigma(F_u)$ *Villarceau spheres* of the tube. The next proposition shows that, as for circles on a torus, they come in two families and are cut out by linear subspaces tangent to the tube.

Proposition 8.3 (Two families)**.** *Let $\rho_\varepsilon(v) = v - 2v_\varepsilon\varepsilon$ be the reflection of $\mathbb{K}$ in $\varepsilon^\perp$. Then:*

(i) *$\Pi_1 \cap T_{R,r} = S_1^+ \cup S_1^-$, where $S_1^\pm$ is the round $(d-1)$-sphere of Π_1 with centre $(\pm r\varepsilon, 0)$ and radius R.*

(ii) *$S_1^- = \sigma(\rho_\varepsilon \times \mathrm{id})(F_1)$ is the image of a fibre of the map $h_\mathbb{K} \circ (\rho_\varepsilon \times \mathrm{id})$, whose fibres on Σ_a also map onto Villarceau spheres partitioning $T_{R,r}$.*

(iii) *$S_1^+ \cap S_1^-$ is the unit $(d-2)$-sphere $\{(x, rx) : x \in \varepsilon^\perp,\ |x| = a\}$, and Π_1 is tangent to $T_{R,r}$ at each of its points.*

The same holds for Π_u, by applying λ_u.

Proof. (i) Write a point of Π_1 as $(x, r\pi(x))$ and put $\xi = x_\varepsilon$, $\eta = |\pi(x)|^2$, so $|x|^2 = \xi^2 + \eta$. Let $G_\pm = (\xi \mp r)^2 + (1+r^2)\eta - (1+r^2)$ be the equations of $S_1^\pm$ in Π_1 (the squared distance from $(\pm r\varepsilon, 0)$ is $(\xi \mp r)^2 + \eta + r^2\eta$). With $\mathcal{A} = \xi^2 + (1+r^2)\eta - 1$ we have $G_\pm = \mathcal{A} \mp 2r\xi$, so $G_+G_- = \mathcal{A}^2 - 4r^2\xi^2$. On the other hand, since $|x| + R > 0$ and $|x|^2 + 1 + r^2\eta > 0$, the equation of the tube restricted to Π_1, $(|x| - R)^2 + r^2\eta = r^2$, is equivalent to $(|x|^2 + 1 + r^2\eta)^2 = 4R^2|x|^2$, that is to

$$(\mathcal{A} + 2)^2 - 4(1+r^2)(\xi^2 + \eta) = 0.$$

Expanding, $(\mathcal{A}+2)^2 - 4(1+r^2)(\xi^2+\eta) = \mathcal{A}^2 - 4r^2\xi^2 + 4\big(\mathcal{A} + 1 - \xi^2 - (1+r^2)\eta\big) = G_+G_-$. So $\Pi_1 \cap T_{R,r} = \{G_+G_- = 0\} = S_1^+ \cup S_1^-$.

(ii) $\rho_\varepsilon \times \mathrm{id}$ is an isometry of $\mathbb{K}^2$ fixing N and preserving Σ_a, and $\sigma \circ (\rho_\varepsilon \times \mathrm{id}) = (\rho_\varepsilon \times \mathrm{id}) \circ \sigma$. Since $\pi \circ \rho_\varepsilon = \pi$, the isometry $\rho_\varepsilon \times \mathrm{id}$ of $\mathbb{R}^d \times \mathbb{R}^{d-1}$ preserves Π_1 and maps the centre $(r\varepsilon, 0)$ to $(-r\varepsilon, 0)$, so it maps S_1^+ to S_1^-. The second statement follows from Theorem E by conjugating with $\rho_\varepsilon \times \mathrm{id}$.

(iii) $G_+ = G_- = 0$ forces $\xi = 0$ and then $\eta = 1/(1+r^2) = a^2$; the points are (x, rx) with $x \in \varepsilon^\perp$, $|x| = a$, and $|(x, rx)|^2 = a^2(1+r^2) = 1$. At such a point p the two spheres $S_1^\pm$ meet transversally inside Π_1, because p is not on the line through their centres ($\xi = 0$ and $\eta \neq 0$), so $T_pS_1^+ + T_pS_1^- = \Pi_1$. Both tangent spaces lie in $T_pT_{R,r}$, hence $\Pi_1 \subset T_pT_{R,r}$. □

For $d = 2$ the reflection ρ_ε with $\varepsilon = i$ is complex conjugation, and the second family consists of the fibres of $(z, w) \mapsto h(\bar z, w)$. For $d = 4, 8$ the two families are the fibres of $h_\mathbb{K}$ and of $h_\mathbb{K} \circ (\rho_\varepsilon \times \mathrm{id})$ on Σ_a; the subspaces Π_u play the role of the bitangent planes, being tangent to the tube along a unit $(d-2)$-sphere.

8.3. **Villarceau's theorem.** Now let $d = 2$, $\varepsilon = i$, and coordinates (x_1, x_2, x_3) on $\mathbb{R}^3 = \mathbb{C} \times \mathbb{R}$. Then $\pi(x) = x_1$, the plane Π_1 is $\{x_3 = r\,x_1\}$, and the circles $S_1^\pm$ have centres $(0, \pm r, 0)$ and radius R.

Proposition 8.4. *Let $T = T_{R,r}$ with $R > r > 0$, not necessarily with $R^2 - r^2 = 1$, and let $\phi \in (0, \pi/2)$ be defined by $\sin\phi = r/R$. The planes through the centre of T that are tangent to T are exactly the planes obtained from*

$$\Pi = \{x_3 = x_1 \tan\phi\}$$

by rotation about the x_3-axis. Each of them is tangent to T at two antipodal points, at distance $\sqrt{R^2 - r^2}$ from the centre, and meets T in the two circles of radius R with centres $(0, \pm r, 0)$ in the case of Π.

Proof. Let Π' be a plane through 0 tangent to T at p. After a rotation about the axis, p lies in the half-plane $\{x_2 = 0,\ x_1 > 0\}$, on the meridian circle $M = \{(x_1 - R)^2 + x_3^2 = r^2\}$. The tangent plane T_pT is spanned by e_2, the direction of the parallel through p, and the tangent line of M at p. As $\Pi' = p + T_pT$ contains 0 and p, its intersection with the plane $x_2 = 0$ is the line through 0 and p, and it is also the tangent line of M at p. The tangent lines of M through the origin are $x_3 = \pm x_1 \tan\phi$, because the distance from the centre $(R, 0)$ of M to such a line is $R\sin\phi = r$.

Their points of tangency are $(R\cos^2\phi, \pm R\cos\phi\sin\phi)$, at distance $R\cos\phi = \sqrt{R^2-r^2}$ from 0. The plane $\{x_3 = -x_1\tan\phi\}$ is the image of Π under the rotation by π about the axis, so Π' is a rotation of Π. Conversely Π contains e_2 and the tangent line of M at $(R\cos^2\phi, 0, R\cos\phi\sin\phi)$, so it is tangent to T there, and by the symmetry $x \mapsto -x$ of T also at the antipodal point.

For the intersection use the coordinates (v, v) on Π, where $x = (v\cos\phi,\ v,\ v\sin\phi)$. The torus is $\{(|x|^2 + R^2 - r^2)^2 = 4R^2(x_1^2 + x_2^2)\}$, and on Π, with $s = v^2 + v^2$,

$$(s+R^2-r^2)^2 - 4R^2(v^2\cos^2\phi + v^2) = \big(v^2+(v-r)^2-R^2\big)\big(v^2+(v+r)^2-R^2\big) + 4v^2\big(R^2\sin^2\phi - r^2\big),$$

an identity of polynomials (Appendix B). The last term vanishes because $\sin\phi = r/R$. So $\Pi \cap T$ is the union of the circles $v^2 + (v \mp r)^2 = R^2$, whose centres are the points $(0, \pm r, 0)$. □

Proof of Corollary 1.1. Let T be a ring torus with radii $R > r > 0$, and let $\mu = \sqrt{R^2 - r^2}$. The similarity $x \mapsto x/\mu$ maps T to $T_{R',r'}$ with $R' = R/\mu$, $r' = r/\mu$ and $R'^2 - r'^2 = 1$, and it maps planes through the centre tangent to T to planes through the centre tangent to $T_{R',r'}$. Put $a = 1/R'$ and $b = r'/R'$; then $a^2 + b^2 = (1 + r'^2)/R'^2 = 1$. By Theorem E and Proposition 8.3, the plane $\Pi_1 = \{x_3 = r'x_1\}$ meets $T_{R',r'}$ in $\sigma(F_1)$ and in its mirror image, the image of a fibre of $h(\bar z, w)$. Here $\tan\phi = r'$ is equivalent to $\sin\phi = r'/R'$ because $R'^2 = 1 + r'^2$, so Π_1 is the plane Π of Proposition 8.4. Every other plane through the centre tangent to $T_{R',r'}$ is $\lambda_u(\Pi_1)$ for a rotation λ_u about the axis, and meets $T_{R',r'}$ in $\sigma(F_u)$ and its mirror image. Scaling back by μ proves the corollary; the circles have radius $\mu R' = R$. □

Remark 8.5. Villarceau's original statement [Vil48] is that the bitangent plane cuts the torus in two circles. Proposition 8.4 proves it directly, and the Hopf fibration explains it: each circle is a great circle of S^3 lying on a flat torus $T_{a,b}$, and stereographic projection maps circles to circles. The geodesics of slope $\pm b/a$ on $T_{a,b}$ (Proposition 7.8) become the two families of Villarceau circles; the meridians and parallels are the images of the coordinate circles $\alpha = \text{const}$ and $\beta = \text{const}$. The Villarceau circles are not the curves $\theta \pm \varphi = \text{const}$ in the usual angular coordinates $(R + r\cos\theta)(\cos\varphi, \sin\varphi)$, $r\sin\theta$ of the torus, which are not circles; they are the curves $\beta \mp \alpha = \text{const}$ in the coordinates of $T_{a,b}$, which differ from the usual angles by a conformal reparametrisation of the meridian.[16]

8.4. **The meridians are Apollonian circles.** As a varies in $(0, 1)$ the tori $T_{R,r} = \sigma(\Sigma_a)$ fill $\mathbb{R}^3$ minus the x_3-axis and the unit circle of the horizontal plane, which are the images of the fibres over the two poles. Their meridians in the half-plane $\{x_2 = 0,\ x_1 > 0\}$ are the circles

$$M_R = \{(x_1 - R)^2 + x_3^2 = R^2 - 1\}, \qquad R > 1.$$

Proposition 8.6. *M_R is the circle of Apollonius $\{P : |P - (1,0)| = \kappa\,|P - (-1,0)|\}$ with $\kappa^2 = (R-1)/(R+1) = (1-a)/(1+a)$. The circles M_R form the hyperbolic pencil of circles with limit points $(\pm1, 0)$, and they are orthogonal to every circle through $(1,0)$ and $(-1,0)$.*

Proof. On M_R we have $x_1^2 + x_3^2 = 2Rx_1 - 1$, so $|P - (1,0)|^2 = 2(R-1)x_1$ and $|P - (-1,0)|^2 = 2(R+1)x_1$. The ratio is $(R-1)/(R+1)$, and $R = 1/a$ gives the second expression. For the orthogonality, an inversion centred at $(-1, 0)$ maps the circles M_R to concentric circles about the image of $(1, 0)$, and the circles through $(\pm1, 0)$ to lines through that point; inversions preserve angles. □

The limit point $(1, 0)$ is the point of the meridian half-plane on the unit circle, the image of the fibre over one pole, and the tori shrink to it as $a \to 1$; as $a \to 0$ they grow towards the axis. Figure 8 shows the pencil together with the bitangent lines of one meridian.

[16] Stereographic projection maps the meridian circle $\{\alpha = \alpha_0\}$ of $T_{a,b}$ onto the meridian of the image torus, but not with constant speed; the two angles differ by a Möbius transformation of the circle.

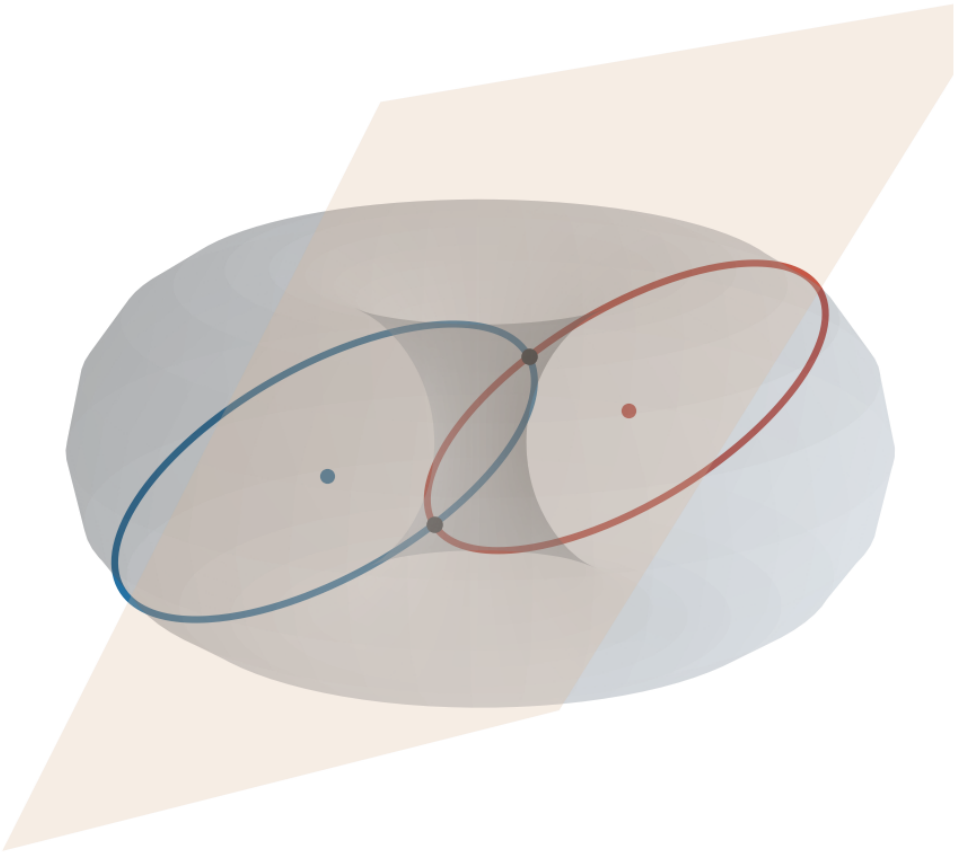


FIGURE 7. A ring torus with $R^2 - r^2 = 1$, the bitangent plane Π through its centre, the two Villarceau circles $\sigma(F_1)$ and its mirror image, of radius R and centres $(0, \pm r, 0)$, and the two points of tangency at distance 1 from the centre.

The point of tangency of every bitangent plane lies at distance $\sqrt{R^2 - r^2} = 1$ from the centre, by Proposition 8.4. So in the meridian half-plane the points of tangency of all the tori $\sigma(\Sigma_a)$ lie on the unit circle, which is the circle through the two limit points orthogonal to all the meridians; in higher dimensions Proposition 8.3(iii) gives the same statement with the unit $(d-2)$-sphere.

9. Stable stems, the image of J and the Kervaire invariant

This section surveys what is known about the stable stems in the range where the Hopf maps are visible. Nothing here is new; the purpose is to state the results precisely, with their sources, and to separate the proved from the announced. Section 9.3 contains the one elementary proof of the section.

9.1. **The stable stems through dimension nineteen.** By Theorem 2.10, $\pi_k^s = \pi_{n+k}(S^n)$ for $n \geq k+2$. Table 2 lists π_k^s for $k \leq 19$, after Toda [Tod62] and Ravenel [Rav86, Table A3.3]; the computations have since been extended far beyond this range by Isaksen, Wang and Xu [IWX23].

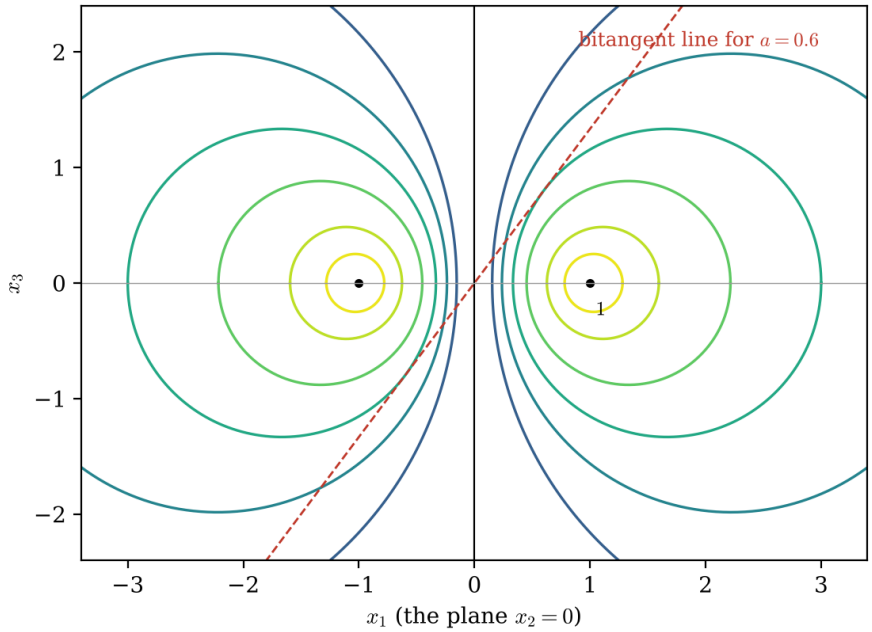


FIGURE 8. Meridians of the tori $\sigma(\Sigma_a)$ in the half-plane $x_2 = 0$: the Apollonian circles with limit points $(\pm 1, 0)$. The line through the origin tangent to one meridian is the trace of a bitangent plane, and its point of tangency lies on the unit circle.

k	0	1	2	3	4	5	6	7	8	9
π_k^s	$\mathbb{Z}$	2	2	24	0	0	2	240	2^2	2^3
k	10	11	12	13	14	15	16	17	18	19
π_k^s	6	504	0	3	2^2	$480\oplus 2$	2^2	2^4	$8\oplus 2$	$264\oplus 2$

TABLE 2. The stable stems π_k^s for $k \le 19$ [Tod62], [Rav86, Table A3.3]; an integer m stands for $\mathbb{Z}/m$ and 2^j for $(\mathbb{Z}/2)^j$.

The groups in degrees 1, 3 and 7 are generated by the suspensions η, ν, σ of the three Hopf maps, and in degrees 2, 6 and 14 the squares η^2, ν^2 and σ^2 are nonzero [Tod62]. The cyclic summands of orders 24, 240, 504, 480 and 264 in degrees $3, 7, 11, 15, 19$ form the image of the J-homomorphism, described next.

9.2. **The J-homomorphism.** An element of $\pi_k(\mathrm{SO}(n))$, represented by $\alpha\colon S^k \to \mathrm{SO}(n)$, defines a map $S^k \times S^{n-1} \to S^{n-1}$, $(x, v) \mapsto \alpha(x)v$, and the Hopf construction turns it into a map $S^{k+n} = S^k * S^{n-1} \to \Sigma S^{n-1} = S^n$. This defines Whitehead's homomorphism $J\colon \pi_k(\mathrm{SO}(n)) \to \pi_{n+k}(S^n)$ [Whi42], compatible with suspension, and in the limit the stable homomorphism

$$J\colon \pi_k(\mathrm{SO}) \to \pi_k^s.$$

For $\mathbb{K} = \mathbb{C}, \mathbb{H}, \mathbb{O}$ the multiplication of unit vectors $S^{d-1} \times S^{d-1} \to S^{d-1}$ is the map associated with the clutching function τ of (3.2), and its Hopf construction is, up to homotopy and sign, the Hopf map $h_{\mathbb{K}}$; this is the classical link between the Hopf maps and J.

By Bott periodicity [Bot59], $\pi_k(\mathrm{SO})$ for $k \geq 1$ depends only on k mod 8:

$k \bmod 8$	0	1	2	3	4	5	6	7
$\pi_k(\mathrm{SO})$	$\mathbb{Z}/2$	$\mathbb{Z}/2$	0	$\mathbb{Z}$	0	0	0	$\mathbb{Z}$

Theorem 9.1 (Adams, Quillen, Sullivan, Becker–Gottlieb)**.** *For $k \geq 1$ the image of J in π^s_{4k-1} is a cyclic direct summand whose order is the denominator of $B_{2k}/4k$, where B_{2k} is the Bernoulli number. For $k \geq 1$ the image of J in π^s_{8k} and in π^s_{8k+1} is $\mathbb{Z}/2$, and J is injective on $\pi_{8k}(\mathrm{SO})$ and $\pi_{8k+1}(\mathrm{SO})$.*

Adams [Ada66] proved the lower bounds and the upper bound for the odd part, and reduced the rest to the *Adams conjecture*: for a real vector bundle ξ over a finite CW complex X and an integer ℓ, there is $e \geq 0$ such that $\ell^e(\psi^\ell \xi - \xi)$ is fibre homotopy trivial, that is, maps to zero in $J(X)$, where ψ^ℓ is the Adams operation. The conjecture was proved by Quillen [Qui71], by Sullivan [Sul74], and by Becker and Gottlieb [BG75].[17]

k	1	2	3	4	5	6	7	8	9	10
stem $4k-1$	3	7	11	15	19	23	27	31	35	39
B_{2k}	$\frac{1}{6}$	$-\frac{1}{30}$	$\frac{1}{42}$	$-\frac{1}{30}$	$\frac{5}{66}$	$-\frac{691}{2730}$	$\frac{7}{6}$	$-\frac{3617}{510}$	$\frac{43867}{798}$	$-\frac{174611}{330}$
$\lvert \mathrm{im}\, J \rvert$	24	240	504	480	264	65520	24	16320	28728	13200

TABLE 3. The order of the image of J in π^s_{4k-1}, the denominator of $B_{2k}/4k$ (Theorem 9.1); the values are recomputed in Appendix C.

Remark 9.2. Write $m(2k)$ for the denominator of $B_{2k}/4k$. By the von Staudt–Clausen theorem the denominator of B_{2k} is the product of the primes p with $(p-1) \mid 2k$, and for the other primes $B_{2k}/2k$ is p-integral (a consequence of the Kummer congruences). Hence

$$m(2k) = 2^{\,2+\nu_2(2k)} \prod_{p \text{ odd},\ (p-1)\mid 2k} p^{\,1+\nu_p(2k)},$$

which gives $m(2) = 2^3 \cdot 3 = 24$, $m(4) = 2^4 \cdot 3 \cdot 5 = 240$, $m(6) = 2^3 \cdot 3^2 \cdot 7 = 504$, $m(12) = 2^4 \cdot 3^2 \cdot 5 \cdot 7 \cdot 13 = 65520$ and $m(14) = 2^3 \cdot 3 = 24$. The formula is checked against Table 3, and against the Bernoulli numbers for $k \leq 60$, in Appendix C.

Figure 9 shows the orders on a logarithmic scale.

9.3. **Division algebras and parallelisable spheres.**

Proposition 9.3. *The spheres S^1, S^3 and S^7 are parallelisable: if $e_1, \dots, e_{d-1}$ is an orthonormal basis of $\mathrm{Im}\,\mathbb{K}$, then $x \mapsto xe_j$, $j = 1, \dots, d-1$, are $d-1$ vector fields on S^{d-1} that are orthonormal at every point.*

Proof. By (D3) and Lemma 3.1, $\langle xa, xb\rangle = \langle a, \bar{x}(xb)\rangle = |x|^2\langle a, b\rangle$. For $|x| = 1$ this gives $\langle xe_j, xe_k\rangle = \delta_{jk}$ and $\langle xe_j, x\rangle = \langle e_j, 1\rangle = 0$, so the vectors xe_j are tangent to S^{d-1} at x and orthonormal. □

The converse is deep: S^{n-1} is parallelisable only for $n = 1, 2, 4, 8$, by Bott–Milnor [BM58] and Kervaire, and Adams determined the maximal number of linearly independent vector fields on every sphere [Ada62]. Together with Corollary 4.6 and Adams' theorem (I4), this is the topological form of the statement that $\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$ are the only real normed division algebras.

[17] The statement of the conjecture in the earlier version of this work was garbled; the formulation here is that of [Ada66].

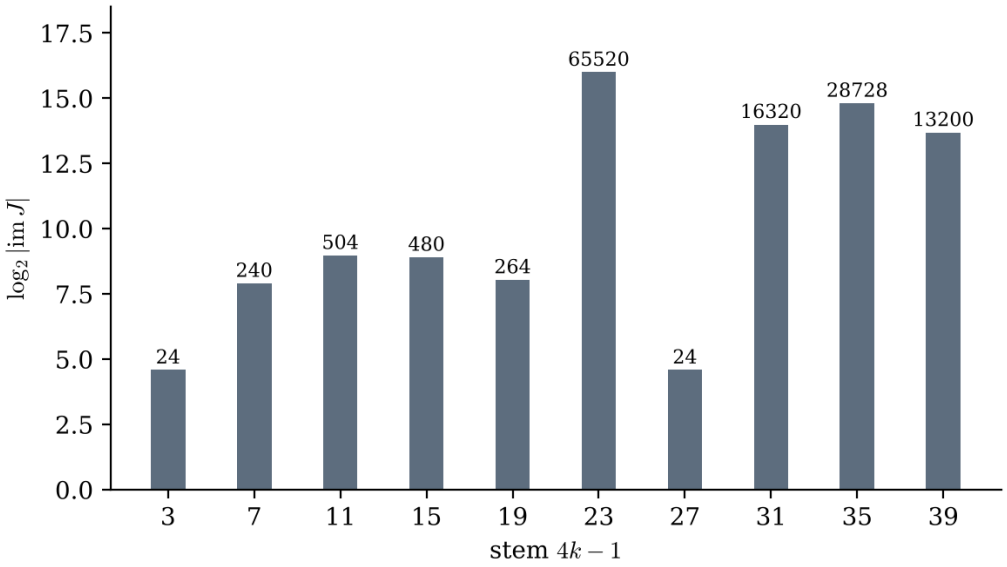


FIGURE 9. $\log_2 |\operatorname{im} J|$ in the stems $4k-1$, $k = 1, \dots, 10$. The order is 24 for $k = 1$ and $k = 7$, when 2 and 3 are the only primes p with $(p-1) \mid 2k$, and it is large when many such primes occur, as for $k = 6$ (the primes $2, 3, 5, 7, 13$).

9.4. **The Kervaire invariant.** A framed manifold of dimension $4j+2$ has a Kervaire invariant in $\mathbb{Z}/2$, the Arf invariant of a quadratic refinement of its intersection form on middle-dimensional homology with $\mathbb{Z}/2$ coefficients; through the Pontryagin–Thom construction it becomes a homomorphism $\pi^s_{4j+2} \to \mathbb{Z}/2$. Browder proved that it vanishes unless the dimension has the form $2^{j+1}-2$ [Bro69]. Let θ_j denote an element of Kervaire invariant one in $\pi^s_{2^{j+1}-2}$, when one exists.

Theorem 9.4. (i) *Elements θ_j exist for $j = 1, 2, 3$, namely η^2, ν^2 and σ^2, in the stems 2, 6 and 14; for $j = 4$ in stem 30* [BMT70]*; and for $j = 5$ in stem 62* [BJM84]*.*
(ii) *No element θ_j exists for $j \geq 7$, that is in the stems $254, 510, \dots$* [HHR16]*.*

The only remaining case is $j = 6$, the stem 126. Lin, Wang and Xu [LWX24a] have announced a proof that θ_6 exists, resting on the machine computations of Adams differentials in [LWX24b]; at the time of writing both papers are preprints, and we record the result as such and do not use it. The first three cases show that the Hopf maps of the three non-commutative division algebras produce, through their squares, the first three elements of Kervaire invariant one.

Remark 9.5. Theorem 9.4 has a geometric meaning: for $j \neq 6$ there is a closed smooth framed manifold of dimension $2^{j+1}-2$ with Kervaire invariant one exactly when $j \leq 5$ [Bro69], [HHR16].

10. Torus fibrations, Calabi–Yau geometry and exotic spheres

The Hopf fibration is the simplest example of several structures that appear in geometry and physics. This section states three of them precisely and proves what is elementary: the Clifford tori are special Lagrangian (Section 10.1), the O'Neill tensor of the Hopf submersions is determined by Theorem F (Section 10.2), and the quaternionic Hopf bundle is the first member of Milnor's family of S^3-bundles over S^4 (Section 10.3).

10.1. **Special Lagrangian tori and the SYZ picture.** Let X be a Calabi–Yau manifold of complex dimension n, with Kähler form ω and holomorphic volume form Ω, with ω^n a constant

multiple of $i^{n^2}\Omega\wedge\bar\Omega$. A real n-dimensional submanifold L is *special Lagrangian* of phase θ if $\omega|_L=0$ and $\operatorname{Im}(e^{-i\theta}\Omega)|_L=0$. The conjecture of Strominger, Yau and Zaslow [SYZ96] proposes that a Calabi–Yau manifold near a large complex structure limit admits a fibration by special Lagrangian n-tori, possibly with singular fibres, and that its mirror is obtained by dualising the tori. The conjecture concerns the existence of such a fibration; it does not prescribe the base.[18]

The model case is the flat Calabi–Yau surface $(\mathbb{C}^*)^2$ with

$$\omega_0=\frac{i}{2}\Big(\frac{dz\wedge d\bar z}{|z|^2}+\frac{dw\wedge d\bar w}{|w|^2}\Big),\qquad \Omega_0=\frac{dz}{z}\wedge\frac{dw}{w},$$

which in the coordinates $z=e^{s_1+i\alpha}$, $w=e^{s_2+i\beta}$ are $\omega_0=ds_1\wedge d\alpha+ds_2\wedge d\beta$ and $\Omega_0=(ds_1+i\,d\alpha)\wedge(ds_2+i\,d\beta)$. This is $(\mathbb{R}\times S^1)^2$ with its flat metric.

Proposition 10.1. *The fibres of* $\operatorname{Log}\colon(\mathbb{C}^*)^2\to\mathbb{R}^2$, $(z,w)\mapsto(\log|z|,\log|w|)$, *are special Lagrangian tori for* (ω_0,Ω_0), *and also Lagrangian for the Euclidean Kähler form of* $\mathbb{C}^2$. *The Clifford tori* $T_{a,b}\subset S^3$ *of Proposition 7.8 are the fibres over the arc* $\{(\log a,\log b):a^2+b^2=1\}$, *and each of them is a union of Hopf fibres, which are the orbits of the diagonal circle in the torus* T^2 *acting on the fibres.*

Proof. On a fibre s_1 and s_2 are constant, so $\omega_0|=0$ and $\Omega_0|=(i\,d\alpha)\wedge(i\,d\beta)=-d\alpha\wedge d\beta$, which is real; so $\operatorname{Im}\Omega_0|=0$. The Euclidean Kähler form is $\frac{i}{2}(dz\wedge d\bar z+dw\wedge d\bar w)=\sum_j\rho_j\,d\rho_j\wedge d\phi_j$ in polar coordinates, and it vanishes on a product of circles. The remaining statements are Proposition 7.8: the Hopf fibres on $T_{a,b}$ are the orbits of $\lambda\mapsto(\lambda z,\lambda w)$. □

Thus the Hopf fibration restricted to $S^3\cap(\mathbb{C}^*)^2$ is the composite of a special Lagrangian torus fibration with the quotient of each torus by a circle. Hopf-like fibrations of Calabi–Yau manifolds in this sense are the subject of [BFS26]; the results there are not used here.

10.2. **The O'Neill tensor of the Hopf submersions.** For a Riemannian submersion $\pi\colon M\to B$ O'Neill [ONe66] defined the tensors T, measuring the second fundamental form of the fibres, and A, measuring the non-integrability of the horizontal distribution; for horizontal vector fields X,Y, $A_XY=\frac12\mathcal{V}[X,Y]$. For orthonormal horizontal X,Y the sectional curvatures satisfy

$$K_B(\pi_*X,\pi_*Y)=K_M(X,Y)+3\,|A_XY|^2. \tag{10.1}$$

Proposition 10.2. *For the Hopf submersions* $S^3\to S^2(\frac12)$, $S^7\to S^4(\frac12)$ *and* $S^{15}\to S^8(\frac12)$ *of Theorem F,* $T=0$ *and* $|A_XY|=1$ *for all orthonormal horizontal vectors* X,Y.

Proof. The fibres are great spheres, which are totally geodesic, so $T=0$. The total space has curvature 1 and the base, a round sphere of radius $\frac12$, has curvature 4, so (10.1) gives $3|A_XY|^2=3$. □

Bounds for the O'Neill tensors of Riemannian submersions in fibred Calabi–Yau manifolds are studied in [GBN26]; Proposition 10.2 is the equality case of the simplest instance, a submersion of a round sphere with totally geodesic fibres.

[18] For a simply connected Calabi–Yau threefold the base is expected to be a homology 3-sphere, and Gross constructed topological T^3-fibrations of the quintic threefold and its mirror over S^3 [Gro01]. This is a theorem about topological fibrations; the existence of special Lagrangian fibrations of compact Calabi–Yau threefolds remains open in general.

10.3. **Milnor's spheres.** For integers h, j let $\xi_{h,j}$ be the real vector bundle of rank 4 over S^4 obtained by clutching two trivial bundles over the hemispheres along

$$f_{h,j}\colon S^3 \to \mathrm{SO}(4), \qquad f_{h,j}(x)(v) = x^h v x^j,$$

and let $M_{h,j}$ be its unit sphere bundle. Milnor [Mil56] showed that $M_{h,j}$ is homeomorphic to S^7 when $h + j = 1$, and that it is not diffeomorphic to S^7 when $(h-j)^2 \not\equiv 1 \pmod 7$; these were the first exotic spheres.

Proposition 10.3. *The bundle $\xi_{1,0}$ is the quaternionic Hopf bundle $L_{\mathbb{H}}$ of Theorem A, so $M_{1,0} = S^7$ with the Hopf map as projection. The maps $x \mapsto x^h$ appearing in $f_{h,j}$ have degree h on S^3 for $h \geq 1$.*

Proof. By Theorem A the clutching function of $L_{\mathbb{H}}$ is $\tau(x) = \ell_x = f_{1,0}(x)$, and $S(L_{\mathbb{H}}) = S^7$ with projection $h_{\mathbb{H}}$. The degree statement is Proposition 4.8. □

So the Hopf fibration $S^3 \to S^7 \to S^4$ is the standard member $(h, j) = (1, 0)$ of the family that contains Milnor's exotic spheres, for instance $(h, j) = (2, -1)$, where $(h-j)^2 - 1 = 8 \not\equiv 0 \pmod 7$. Hopf-like fibrations of exotic spheres are studied in [Bha26b].

Appendix A. The octonions

We use the Cayley–Dickson model $\mathbb{O} = \mathbb{H} \oplus \mathbb{H}$ with the product (3.1), and the basis $e_0 = 1$, $e_1 = i$, $e_2 = j$, $e_3 = k$ of the first summand and $e_{4+m} = (0, e_m)$, $m = 0, 1, 2, 3$, of the second. Table 4 is the resulting multiplication table, computed from (3.1). Equivalently, $e_a^2 = -1$ for $a \geq 1$, $e_a e_b = -e_b e_a$ for $a \neq b$, and $e_a e_b = e_c$ (together with its cyclic permutations) for the seven oriented triples

$$(1,2,3),\ (1,4,5),\ (1,7,6),\ (2,4,6),\ (2,5,7),\ (3,4,7),\ (3,6,5),$$

which are the lines of the Fano plane on the points $1, \dots, 7$.

$\cdot$	e_1	e_2	e_3	e_4	e_5	e_6	e_7
e_1	-1	e_3	$-e_2$	e_5	$-e_4$	$-e_7$	e_6
e_2	$-e_3$	-1	e_1	e_6	e_7	$-e_4$	$-e_5$
e_3	e_2	$-e_1$	-1	e_7	$-e_6$	e_5	$-e_4$
e_4	$-e_5$	$-e_6$	$-e_7$	-1	e_1	e_2	e_3
e_5	e_4	$-e_7$	e_6	$-e_1$	-1	$-e_3$	e_2
e_6	e_7	e_4	$-e_5$	$-e_2$	e_3	-1	$-e_1$
e_7	$-e_6$	e_5	e_4	$-e_3$	$-e_2$	e_1	-1

Table 4. The product $e_a e_b$ (row a, column b) in the Cayley–Dickson basis of $\mathbb{O}$.

Proposition A.1. *Let A be an associative real algebra with involution in which $x\bar{x} = \bar{x}x = |x|^2$ is a positive definite multiplicative quadratic form and $\mathrm{Re}(xy) = \mathrm{Re}(yx)$, where $\mathrm{Re}\, x = \frac{1}{2}(x + \bar{x})$. Then the double $A \oplus A$ with the product (3.1) has the same properties except possibly associativity; in particular $|xy| = |x||y|$ on $\mathbb{H} \oplus \mathbb{H} = \mathbb{O}$.*

Proof. Let $x = (a, b)$ and $y = (c, d)$. Then $xy = (ac - \bar{d}b,\ da + b\bar{c})$ and

$$|xy|^2 = |a|^2|c|^2 + |d|^2|b|^2 - 2\,\mathrm{Re}\big(ac\,\bar{b}d\big) + |d|^2|a|^2 + |b|^2|c|^2 + 2\,\mathrm{Re}\big(da\,c\bar{b}\big),$$

using $\overline{\bar{d}b} = \bar{b}d$ and $\overline{b\bar{c}} = c\bar{b}$. The two cross terms cancel because $\mathrm{Re}(d \cdot ac\bar{b}) = \mathrm{Re}(ac\bar{b} \cdot d)$, so $|xy|^2 = (|a|^2 + |b|^2)(|c|^2 + |d|^2) = |x|^2|y|^2$. The remaining properties are checked in the same way. □

Applied to $A = \mathbb{C}$ and $A = \mathbb{H}$, the proposition gives (D1) for $\mathbb{H}$ and $\mathbb{O}$. The identities (D2) and (D3) follow from the definitions in the same manner [Sch66, Chapter III]. Artin's theorem (D4) is proved in [Sch66, Theorem 3.1]; for $\mathbb{O}$ it is equivalent to the alternative laws $(xx)y = x(xy)$ and $(yx)x = y(xx)$.

The octonion identities used in the text are:

(O1) $(xy)y^{-1} = x = (xy^{-1})y$ and $x^{-1}(xy) = y$ (Lemma 3.1), used for the bundle charts in Section 3;
(O2) $(xc)\bar{c} = |c|^2 x$, used to identify the bundle projection with $h_{\mathbb{O}}$;
(O3) $\langle xa, xb\rangle = |x|^2\langle a, b\rangle$, used for the parallelisations of Proposition 9.3 and the isometries Λ_u of Section 8;
(O4) $wx = \bar{x}w$ for $w \perp 1$, $w \perp u$, $x \in \mathbb{C}_u$, used in Lemma 4.7.

All of them involve at most two independent octonions, which is why they hold although $\mathbb{O}$ is not associative. An identity that involves three octonions can fail: $(e_1e_4)(e_2e_4)^{-1} = e_3$ but $e_1e_2^{-1} = -e_3$, which is the explicit triple of Proposition 3.7.

Appendix B. The algebra of Villarceau's theorem

B.1. **The image of a Hopf fibre.** For $d = 2$, $0 < a < 1$, $b = \sqrt{1-a^2}$, the base fibre is (ae^{it}, be^{it}) and its stereographic image is

$$X(t) = \frac{\big(a\cos t,\ a\sin t,\ b\cos t\big)}{1 - b\sin t}.$$

Its third coordinate is (b/a) times its first, so it lies in the plane $x_3 = (b/a)x_1$. With $\varsigma = \sin t$, the vector $a(1-b\varsigma)\big(X(t) - (0, \frac{b}{a}, 0)\big)$ is

$$\big(a^2\cos t,\ a^2\varsigma - b(1-b\varsigma),\ ab\cos t\big) = \big(a^2\cos t,\ \varsigma - b,\ ab\cos t\big),$$

using $a^2 + b^2 = 1$, and its squared length minus $(1-b\varsigma)^2$ is

$$a^2(a^2+b^2)\cos^2 t + (\varsigma - b)^2 - (1-b\varsigma)^2 = a^2(1-\varsigma^2) + (1-b^2)(\varsigma^2 - 1) = 0.$$

So $|X(t) - (0, \frac{b}{a}, 0)| = 1/a = R$. This is the case $d = 2$ of Proposition 8.2.

B.2. **The bitangent factorisation.** Let $x = (\upsilon\cos\phi,\ v,\ \upsilon\sin\phi)$ and $s = \upsilon^2 + v^2 = |x|^2$. Then

$$\begin{aligned}
&(s + R^2 - r^2)^2 - 4R^2(\upsilon^2\cos^2\phi + v^2)\\
&\quad= \big(s + r^2 - R^2\big)^2 - 4r^2v^2 + 4s(R^2 - r^2) - 4R^2\upsilon^2\cos^2\phi - 4(R^2 - r^2)v^2\\
&\quad= \big(\upsilon^2 + (v-r)^2 - R^2\big)\big(\upsilon^2 + (v+r)^2 - R^2\big) + 4\upsilon^2\big(R^2 - r^2 - R^2\cos^2\phi\big),
\end{aligned}$$

where the first step uses $(s + R^2 - r^2)^2 - (s + r^2 - R^2)^2 = 4s(R^2 - r^2)$ and the second uses $(s + r^2 - R^2)^2 - 4r^2v^2 = \big(s + r^2 - R^2 - 2rv\big)\big(s + r^2 - R^2 + 2rv\big)$. Since $R^2 - r^2 - R^2\cos^2\phi = R^2\sin^2\phi - r^2$, this is the identity used in Proposition 8.4.

B.3. **The higher-dimensional factorisation.** In the notation of Proposition 8.3, with $\mathcal{A} = \xi^2 + (1+r^2)\eta - 1$ and $G_\pm = \mathcal{A} \mp 2r\xi$,

$$(\mathcal{A} + 2)^2 - 4(1+r^2)(\xi^2 + \eta) = \mathcal{A}^2 - 4r^2\xi^2 = G_+G_-,$$

because $4\mathcal{A} + 4 - 4(1+r^2)\xi^2 - 4(1+r^2)\eta + 4r^2\xi^2 = 4\big(\mathcal{A} + 1 - \xi^2 - (1+r^2)\eta\big) = 0$. For $d = 2$, $\xi = x_2$ and $\eta = x_1^2$, and the identity reduces to the previous one with $\tan\phi = r$.

Appendix C. Verification programs

Every statement of this paper is proved in the text or quoted from the literature, and no proof depends on a computer. The programs described here, which accompany the paper in the repository named in the Data availability statement, recompute independently the finite identities and the numerical values that appear in the text. The program `code/checks.py` uses NumPy and SymPy and runs in about twenty seconds; it ends with the line `58 checks passed, 0 failed`. It checks:

(1) the octonion identities: non-associativity, the alternative laws, the multiplicativity of the norm, the identities of Lemma 3.1, the Moufang identity $(xy)^{-1} = y^{-1}x^{-1}$, and the triple (e_1, e_2, e_4) of Proposition 3.7;
(2) for $d = 2, 4, 8$, that $|h_{\mathbb{K}}| = 1$ on the unit sphere, that $h_{\mathbb{K}}$ is constant on each subspace L_x, and that $h_{\mathbb{K}}(L_x)$ is the stereographic image of x (Theorem A);
(3) the identity $h(u) \cdot h(v) = 2|\langle u, v\rangle|^2 - 1$ for $\mathbb{C}$ and $\mathbb{H}$ (Proposition 7.2);
(4) for $d = 2, 4, 8$, that $Dh_{\mathbb{K}}$ has d singular values 2 and $d - 1$ zeros on the tangent space of S^{2d-1}, at random points (Proposition 7.5), and the decomposition of Lemma 7.4;
(5) for $d = 4, 8$ and $m = 2, 3, 5$, the roots of $x^m = y$ and the signs of the Jacobian at them, and the derivative formula of Lemma 4.7;
(6) Gauss integrals (6.1): two Hopf fibres link once, and two fibres of $[z^p : w^q]$ link pq times for six pairs (p, q);
(7) the Hopf invariants $2, 0, 0, 4, 6$ of the maps $[z^2 - \frac{1}{4} : w]$, $[z^2 - 4 : w]$, $[z^2 : zw + 1]$, $[z^2 + w^2 : zw]$ and $[z^2 : w^3]$ of Example 6.7, computed as degrees by integrating the pull-back of the volume form of S^3;
(8) the degrees 2, 6, 0 of $F/|F|$ for the split quaternionic maps $(a^2 + c, b)$ and $(a^2 + c, b^3 + c/2)$ with a small non-real c, and $(a^2 + c', b^3 + c')$ with a large one (Theorem 6.15), as signed counts of the preimages of a regular value;
(9) the Jacobian formula (6.4) for random one-sided cubics over $\mathbb{H}$ and $\mathbb{O}$, and the zeros $x + yJ$, $J = -UV^{-1}$, read off from the roots of S_P (Lemma 6.12);
(10) for the map of Example 6.16, two preimages of a small regular value in the unit ball, both with positive Jacobian;
(11) the conjugation formula (6.8) and the determinant -1 of conjugation;
(12) the Villarceau identities of Appendix B, symbolically, and Theorem E for $d = 4, 8$ numerically: the images of the fibres F_u are spheres of radius R about $(ur\varepsilon, 0)$ in Π_u, on the tube, and the mirror family gives the spheres about $(-r\varepsilon, 0)$;
(13) the denominators of $B_{2k}/4k$ in Table 3, and the formula of Remark 9.2 for $k \le 60$;
(14) the consistency of Table 1 with Theorem C and with Freudenthal's theorem, as abelian groups.

The figures are produced by `figures/make_figures.py` with Matplotlib.

Declarations

Ethics statement. This article is a work of pure mathematics. It involves no studies with human participants or animals and no personal data, and it required no approval from an ethics committee.

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Data availability. No data were generated or analysed. The verification programs of Appendix C and the programs that produce the figures are available in the repository `https://github.com/creelie/Paper_6`. AI-assisted software was used to write and run these programs and to check computations; every statement and proof was checked by the authors, who take full responsibility for the content.

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